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His Depression, Her Depression: A Corpus-Based Analysis of
Gendered Linguistic Markers of Depression

Jego Depresja, Jej Depresja: Analiza Korpusowa Językowych Markerów Depresji w
Perspektywie Płci

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Introduction

The present thesis lies at the intersection of three scientific fields: computational linguistics, mental health research, and gender studies. Its aim is to investigate whether the language used by individuals with depression differs systematically from that of healthy controls, and whether any such differences are shaped by the speaker's sex/gender. These questions are situated within the broader context of Affective Computing, a rapidly developing field concerned with 'teaching' computers how to recognise and process human emotional states, which has increasingly turned its attention towards the automated detection of mental health conditions.

Depression is one of the most prevalent mental health disorders worldwide, affecting approximately 332 million people globally (www.who.int/news-room/fact-sheets/detail/depression, DOA: 15.01.2026). Despite being present within the medical canon since the Hippocratic times, and extensive diagnostic manual guidelines, its diagnosis remains largely dependent on subjective self-report and practitioner judgement. The growing availability of digital corpora and advances in Natural Language Processing have opened new possibilities for objective, data-driven approaches to depression detection; such possibilities have been keenly examined and developed by researchers, with the aim of facilitating the diagnostic process. Language, as the most direct and reliable expression of human thought and emotion, represents a promising source of diagnostic information. A substantial body of empirical research has identified multiple linguistic features as markers of depressive cognition; among these three features, negative affect words, first person singular pronouns, and absolutist language are some of the most prevalent. However, the majority of this research has been conducted on written text from anonymous sources, e.g. online fora, and relatively little attention has been paid to the role of gender in shaping these linguistic patterns.

The present study addresses both of these gaps by conducting a corpus-based analysis of spoken interview transcripts from the Distress Analysis Interview Corpus — Wizard of Oz (DAIC-WOZ), a dataset of semi-structured clinical interviews designed to facilitate automated depression assessment. Using the Linguistic Inquiry and Word Count (LIWC) software, a set of theoretically motivated linguistic markers is examined across depressed and non-depressed participants, and subsequently across depressed men and depressed women, to determine whether gender moderates the linguistic expression of depression.

Two research questions guide the study:

RQ1: Do symptoms of depression manifest themselves in patient language in the form of linguistic markers?

RQ2: Do differences in the epidemiology, phenomenology, and pathophysiology of depression between men and women translate to differences in linguistic marker presence?

These questions give rise to four hypotheses. The broad hypothesis that depressed and non-depressed participants will differ in their language use (H1) is operationalised through three sub-hypotheses targeting specific linguistic markers: negative emotional valence (H1a), first person singular pronoun use (H1b), and absolutist word frequency (H1c). A second hypothesis (H2) predicts that depressed men and women will differ in their use of linguistic markers of depression.

The thesis consists of three chapters. Chapter One provides the theoretical background of the study. It introduces the fields of Natural Language Processing and Corpus Linguistics, traces the development of Affective Computing from its early goals to its contemporary clinical applications, and presents an overview of mental health research with particular attention to the history, diagnosis, and treatment of depression. The chapter then examines the links between language and mental health, reviewing both nonverbal and text-based markers of depression identified in the literature. It concludes by addressing the relationship between gender, language, and depression through the lens of neurofeminist research, with specific attention to the Silencing the Self theory as a framework for understanding gendered emotional expression.

Chapter Two explains the methodology of the study. It describes the DAIC-WOZ corpus in detail, including its participant profile and interview protocol, and explains the PHQ-8 psychological instrument used to classify participants as depressed or non-depressed. The chapter then presents the research questions and hypotheses, after which it describes the LIWC software and provides theoretical and empirical justification for the selection of linguistic markers. It concludes with a step-by-step account of the database compilation and pre-processing procedure.

Chapter Three presents and discusses the results of the statistical analyses. For RQ1, inferential group comparisons and logistic regression are employed. For RQ2, inferential group comparisons are conducted on the depressed sub-sample, followed by exploratory logistic regression models run separately for all male and all female participants. The results

are discussed in relation to prior literature, theoretical frameworks, and the broader implications for automated depression detection and gender-sensitive clinical tools. The chapter concludes with an overview of the study's limitations and proposes directions for future research.

Chapter One

Theoretical Background

Introduction

This chapter establishes the theoretical and contextual foundations necessary to situate the present research within its broader academic perspective. Within it, I provide the background on six interconnected research fields, which together present the interconnected landscape of computational linguistics, affective science, mental health research, and gender studies.

The chapter begins with an introduction to the branch of Natural Language Processing, as well as its emerging sub-tasks of Natural Language Understanding and Natural Language Generation, taking into consideration the impact of Artificial Intelligence research. I then proceed to present the interconnected backdrop of Corpus Linguistics, the scientific methodology on which this thesis relies. This is followed by an overview of Affective Computing, tracing the field from its original objectives and early applications through to its contemporary goals.

The discussion then turns to Mental Health Research, depicting the history of approaches to mental health worldwide, and presenting the contemporary state of the field with its future directions in mind. Particular attention is paid to depression specifically, examining its frameworks as a clinical category through time, and analysing the diagnosis criteria present in current manuals, such as the DSM-5 and the ICD-11.

Section 5 of the chapter attempts to link the information established throughout the previous sections, to infer the links between linguistic behaviour, mental health, and gender. Specifically, the section presents both verbal and non-verbal markers of depression, and introduces the idea of linguistic differences between the genders. The chapter concludes by examining these themes through the lens of neurofeminism, addressing sex-based differences in depression and the relevance of the Silencing the Self theoretical framework.

Together, these sections provide the conceptual foundation on which the empirical work of this thesis builds.

1. Natural Language Processing

Natural Language Processing (NLP) first emerged in the mid-20th century, as an interdisciplinary field between linguistics and artificial intelligence (Nadkarni et al., 2011). Currently, it also stands alone as its own scientific field, functioning under the broader name of “computational linguistics”. NLP can be described as “a theoretically motivated range of computational techniques for analyzing and representing naturally occurring texts at one or more levels of linguistic analysis for the purpose of achieving human-like language processing for a range of tasks or applications” (Liddy, 2001, pp. 3). Put simply, the early goals of NLP were to enable computers to recognise the rules of natural language, both speech and text, for the purpose of computing linguistic analyses. While understanding language structure may be an intuitive task for humans, “teaching” these skills to computer programmes was more challenging; in order to successfully parse sentences, a computer must be able to: know where one word ends and the next starts; differentiate between parts of speech (POS), i.e. nouns and verbs; recognise which sequences of words are acceptable, and which are not; and much more (Hirschberg and Manning, 2015). Early advances in NLP created many possibilities, with the field enabling tasks such as machine translation, the creation of conversational computer agents, speech recognition, or sentiment analysis.

Contemporarily, an even more complex sub-field of NLP has arisen: Natural Language Understanding (NLU). As can be inferred from this subtasks name, NLU aims to go one step further than NLP, not only enabling computers to process the syntactic structure of human language, but also to understand the semantic content being processed. NLU is closely interlinked with Natural Language Generation (NLG), the process of computers generating answers in human language based on received prompts. The combined goals of NLU and NLG have been described as follows: “[b]y addressing lexical, syntactical and referential ambiguities, the focus has been to extract intention, i.e. the purpose of a certain text, to acknowledge entities, i.e. the entities that are involved in the passage, and to characterize emotions that are hidden behind the text” (Karanikolas et al., 2023, pp. 279). NLU and NLG have drastically benefitted from advances in Artificial Intelligence (AI) research, especially from the development of complex Large Language Models (LLM). Based on state of the art Transformers, LLMs leverage artificial neural network architecture to perform demanding linguistic tasks with unprecedented proficiency (Ferraris et al., 2024). Karanikolas et al. (2023) provide the following explanation of LLM functioning:

After being fed with enormous and of great variety text data sources, the LLM produces a trained model. Its knowledge is transformed into Embeddings, i.e. numerical representations of words in the form of vectors, that capture the words' semantics, contextual relationships, and the different kinds of meanings in different contexts. When the system is introduced to a new user prompt, this prompt goes under the same transformation. Trying to respond to the user, the system calculates vector similarity functions to estimate which response best describes the user's needs, and finally decodes it to a human-like output. (pp. 281)

LLM technology has been behind the boom of AI chatbots, such as ChatGPT, Gemini, and others. It is the introduction of LLMs which introduced NLP to the public eye, spreading computational linguistics tasks into a vast array of fields, from business and marketing, to medical and legal research. For a more detailed explanation of LLM and Transformer technology, see Ferraris et al. (2024).

1.1. Corpus Linguistics

Corpus linguistics and NLP research have been mutually developing over the past decades. The notion of a corpus, which the field is named after, can be loosely defined as a “large and principled collection of texts” (Biber et al., 1998, pp. 4). More specifically, McEnery and Wilson (1996) specify four characteristics and recommendations which distinguish corpora from any other collected bodies of text: a corpus’ sample should aim to be maximally representative of an examined language variety; a corpus is of a finite size, which can either be closed, or open to new additions; a corpus should be compiled in a machine-readable form, meaning that it can be processed by means of NLP tools; a corpus’ sample should be generalisable. Hence, corpus linguistics utilises compiled corpora of speech or text to conduct statistical linguistic analyses, so as to make assumptions or predictions about a wider population or language variety. Gries (2009) further specifies, that the backbone of corpus linguistics analyses are frequencies; by assessing the frequencies of occurrence of linguistic elements, or frequencies of these elements’ co-occurrences, linguists can make “conceptual leaps” (pp. 1226) to infer information about cognitive processes. Corpora are useful analytical tools, as they allow for both qualitative and quantitative approaches; while the latter is more prevalent, corpora can also be manually annotated according to coding schemes crafted for the research questions at hand.

The advances of technology and the Internet have also revolutionised how corpus linguistics research is done. On the one hand, corpora can be compiled through manual data collection, such as interviewing participants, as is the case for the corpus on which this thesis is based on (see Chapter Two, Section 1). On the other hand, increasingly often corpora are compiled from the masses of text available online. The ready availability of data has

increased the scope of corpus analyses, as the most time-consuming step of data collection has been reduced. Additionally, Holtz et al. (2012, pp. 55) highlight that “[f]or radical, extremist, and other ideologically “sensitive” groups and organizations in particular, Internet forums are a very efficient and widely used tool to connect members, inform others about the group’s agenda, and attract new members”. This means that analyses on sensitive topics can be conducted, without the need of impacting ecological validity in experimental conditions. By combining this data availability with the increased proficiency of NLU and LLM analyses, computational linguistics has extended its reach and become a cornerstone of linguistic research.

2. Affective Computing

Affective Computing (AffC) is the term coined by Rosalind Picard in 1997, used to describe the scientific field concerned with programming computers and machines to recognise, process, express, and regulate human emotions (Picard, 1997). Introduced at a time when computers had started to become regular household items, yet were still novel to the everyday users, AffC was a revolutionary field which brought attention to core aspects of how both computers and humans functioned. As highlighted many times in Picard’s *Affective Computing* (1997), being “emotional” often had negative connotations to the public eye; emotions were thought to cloud rational thinking and to detrimentally influence decision-making and other “logical” processes. Hence, to many, aiming to “give” computers emotional abilities may have seemed to be an unprofitable endeavour; Picard, however, noticed many promising possibilities for the future of computer science, stating that “emotions are an active part of intelligence, especially perception, rational thinking, decision making, planning, and creativity” (Picard, 1997, pp. 247). Now, AffC is a pioneering sub-task of computer science.

The process of teaching machines how to “feel”, or even simply how to understand feelings, is made difficult by the fact that scientists still do not fully grasp the inner workings of emotions in humans. Moors (2024) alone lists five main schools of emotion theories (evolutionary theories, network theories, evaluation theories, psychological constructionist theories, and social theories), and many more sub-schools exist. In an attempt to assess the state of emotion theories, Lazarus posed twelve questions about emotions, which a functioning emotion theory should be able to tackle. These are as follows:

1. How should emotion be defined?

2. What constitutes being an emotion?
3. Should emotions be seen as discrete or continuous categories?
4. How is physiological activity linked to emotion?
5. How are the different emotions interdependent?
6. How are cognition, motivation, and emotion connected?
7. How can the evolutionary and sociocultural aspects of emotion be connected and explained?
8. Do consciousness and appraisal affect emotion?
9. How is each emotion generated?
10. How should emotional development be understood?
11. How do emotions impact the human state over time?
12. How can emotions be dealt with and managed? (Lazarus, 1991, pp. 34–35).

While the list highlights what is important when thinking about emotions, it does not necessarily provide clarity in the field; of the five previously mentioned theories, each postulates its own terms of constitution, causality, and mechanism. A computer scientist attempting to work in the field of affective computing must first decide what definition of emotion and feeling they stand by. Apart from this set of questions, many more arise in the field of Affective Computing itself. For example, Picard is clear about the benefits of machines understanding emotions, however she deliberates about how this should be done, for what reasons it should be done, and, importantly, how these emotional skills can be tested in a machine (1997, pp. 69–70). It is due to these many questions that the goals, tasks, and applications of AffC have varied widely over time.

2.1. The Early Goals and Applications of Affective Computing

At the early stages of introducing AffC, Picard focused on presenting what conditions a computer would have to satisfy to possess “emotional intelligence”, which is defined as “the ability to recognize, express, and have emotions, coupled with the ability to regulate these emotions, harness them for constructive purposes, and skillfully handle the emotions of others” (1997, pp. 49). This definition of recognition highlights the importance of three main emotional abilities: recognition, expression, and possession. Picard provided criteria for each of these skills, all of which should be satisfied by an emotionally intelligent computer. Each of these skills is theoretically modelled to match the human cognitive-emotional process, however, space is left for computers to evolve in such a way that they develop differently from their human prototypes.

The ability to recognise human emotions can be seen as the most necessary of Picard's emotional intelligence criteria, as it creates the baseline for human-computer interaction; emotion recognition requires inference, context-sensitivity, and interpretation of facial expressions, gestures, tone, and textual content. A computer which can recognise emotion is able to elevate communication from mere semantic understanding, to pragmatic understanding. However, it is important to note that in order to fully recognise human emotions, a computer must be able to fulfill all of the other emotional intelligence tasks (Picard, 1997, pp. 50–53). Picard further suggested that computers may surpass human ability, being able to recognise affective states which humans have not yet named or identified; this is due to the fact that an affective computer is able to receive more information than a human counterpart, such as electrodermal responses, brainwaves, or blood pressure, alongside “seeing” facial expressions, gestures, and more (1997, pp. 53). However, to simply match human ability, a computer capable of recognising human emotions must be capable of: 1a) receiving a variety of input signals; 1b) recognising patterns, e.g. vocal intonations; 1c) reasoning and predicting which affective state is being exhibited, while taking context, past and present situations, personal preferences, and social rules into consideration; 1d) learning over time to recognise patterns which vary between users, which leads to future optimisation; 1e) possessing its own biases (just as a human may be more or less biased towards different people and situations); 1f) providing an output, consisting of the predicted recognised emotions (Picard, 1997, pp. 55). Only when these criteria are fulfilled can a computer be said to be capable of recognising emotion.

A different set of criteria is presented for computers to be able to express emotions. In this case, Picard specified that a computer must be capable of: 2a) receiving input from a human, different computer, or an in-built emotion generation algorithm, which will provide information about which emotion should be expressed; 2b) having two emotion generation pathways, one of which is spontaneous and uncontrolled, and the other of which is intentional; 2c) displaying feedback, which impacts its expression and affective state; 2d) excluding bias, meaning that a particular affective state may make it more difficult for the computer to return to the previously expressed states; 2e) knowing about social display rules and being aware of social norms; 2f) returning some output, which can take many forms, e.g. avatars with facial and bodily expressions, synthesised speech, or others (Picard, 1997, pp. 60). Such skills would serve as the basis for elevated human-computer interaction, as a computer would not only recognise what its interlocutor was feeling, but would also have the ability to answer in an affective manner. Moreover, these criteria also provide room for the

development of advanced, currently unexplored, levels of computer-computer interaction, as point 2a) clearly specifies that the initial emotional input does not need to come from a human interlocutor. Affective computers may develop in a way which has not yet been considered.

While expressing emotions may seem like an end goal in the field of AffC, emotion theorists and computer scientists highlight that emotion expression is not equal to emotion possession. For this reason, Picard outlined a final set of criteria, specific to the skill of emotion possession: 3a) having behaviour which results from experiencing emotions; 3b) having spontaneous primary emotion responses to certain inputs; 3c) cognitively generating emotions by reasoning about hypothetical possibilities; 3d) having cognitive awareness, physiological awareness, and subjective feelings; 3e) being subject to the effects of emotion on other system functions, such as memory, perception, decision making, learning, etc. (Picard, 1997, pp. 70). To test and prove this final stage of a machine's capacity for emotions, Picard proposed a new take on the well-known Turing test. To successfully pass this particular test, it is not enough for the machine to simply have a conversation with a human. Instead, the computer “would need to be capable of recognizing emotion and synthesizing a suitable affective response” (Picard, 1997, pp. 13). The conversation between human and computer would need to revolve around emotional topics, and both sides would need to be prodded and provoked to exhibit realistic affective responses.

In her seminal work, apart from presenting guidelines for how Affective Computing should be achieved, Picard proposed possible future applications of this newly-developing field. The provided examples belong to the fields of medicine, marketing, consumer behaviour, education, and theoretical sciences. In terms of medicine, it was proposed that AffC could improve the quality of text-to-speech synthesizers used by individuals with speech impediments, be a helpful tool in therapy, or benefit individuals in the autism spectrum who have a difficulty with recognising emotions in themselves and in others (Picard, 1997, pp. 88–90). For marketing purposes, AffC could be used to measure customer feedback more accurately, e.g. by sensing responses to video games, music, or films, ultimately benefitting marketers (Picard, 1997, pp. 92, 107–108). However, AffC can simultaneously benefit consumers, as Picard provided suggestions on how affective computers would be better at learning user preferences, e.g. matching the music choice to the listener's mood, knowing which parts of a video to skip etc., possibly predicting a consumer's needs better than a consumer themselves could (1997, pp. 98–99). From the perspective of education, Picard noted how adaptive feedback had been found to improve learning abilities

of students and could be built upon with AffC tools, and also deliberated about how teachers could benefit if AffC was used to assess the emotional states present within the classroom, making it easier for them to adapt to the students' emotional needs (1997, pp. 93–94). While not all of Picard's goals have been realised, her theoretical and practical work provided the field of Affective Computing with a solid foundation for future scientists to work upon and develop further.

2.2. Current Goals and Applications of Affective Computing

Due to rapid technological and scientific developments, Affective Computing has developed drastically over the three decades since its introduction in the late 20th century. Benefitting from innovations such as the evolution of Machine Learning (ML) and Deep Learning (DL), as well as the development of Extended Reality technology (XR), which includes the likes of Virtual Reality (VR), Augmented Reality (AR), and Mixed Reality (MR), Affective Computing has made significant progress on achieving the goals outlined by Picard (1997) and other early representatives of the scientific field. The growing significance of this field of research is quantifiable through the number of papers published on the topic, reaching 33,448 published articles as of 2023 (Pei et al., 2024).

As more research has been done on Affective Computing, the tasks and future goals of the academic discipline have undergone changes. Contemporarily, the content of AffC research can be divided into five subfields: fundamental theory, emotion generation and expression, signal collection, multimodal fusion, and sentiment analysis (Pei et al., 2024). Each AffC sub-task is tailored towards achieving different results. Fundamental theory is closely linked to studies in psychology, which task themselves with building models of emotion. Moreover, Fundamental Theory is concerned with building on the theoretical background put forth in Picard's *Affective Computing* (1997), strengthening the knowledge possessed by researchers in this field. In contrast, the remaining sub-tasks are of a more practical nature.

The goal of emotion generation and expression is closely linked to contemporary researchers attempting to implement the theoretical work into practical products. An example of this are the Generative Pre-Trained Transformer (GPT) models, created by OpenAI, which take the format of conversational chatbots. An improvement made upon the GPT-2 model was suggested to be capable of “generating affect-driven and topic focused sentences without

losing grammatical correctness as the affect intensity increases” (Goswamy et al., 2020, pp. 2787). Research on newer model versions, such as GPT-3.5-Turbo and GPT-4o has shown that these auto-regressive models perform exceptionally well at emotion detection, when compared to other widely available models (Lecourt et al., 2025, pp. 2609). The newest advancement which reflects the early reflections of AffC theory, is the introduction of “personality” in ChatGPT 5 model (help.openai.com/en/articles/11899719-customizing-your-chatgpt-personality, DOA. 25.02.2026). A user can now choose from a list of personalities for their chatbot assistant to adopt, so as to better suit their conversational style and needs. The personalities range from the serious, such as “professional”, or “candid”, to the informal, like “friendly”, or “quirky”. While computers may still be far from “having” emotions according to Picard’s (1997) definition, they are now certainly capable of expressing them during conversation with humans, improving the quality of human-computer interaction.

The task of signal collection is less related to the readily available end product, in this case being an affective computer, but instead more so a subtask for helping develop the technology behind affective software. To make models such as GPT possible, open-access databases containing textual, speech/audio, visual, physiological, or multimodal data must be made available for training the models in the first place (Pei et al., 2024). As with emotion generation, the rapid technological progress has made the task of collecting raw data for further analysis has become increasingly easy; for example, large amounts of textual data have become readily available as a result of the development of social media. In a systematic review of AffC assets, Vaz et al. (2025, pp. 540) found as many as 111 different datasets of varying modalities available for affectively-intelligent model training. Among the notable datasets, one can find the IMDB database containing thousands of emotive reviews, the JAFFE database containing several images of a range of facial expressions, or the bimodal FAcE and BOdy database (FABO), which provides both facial and bodily gestures (Wang et al., 2022, pp. 24). Many more datasets are available, and numbers will presumably continue to increase as AffC steadily grows in popularity.

The databases are of utmost importance for the remaining two tasks of AffC: multimodal fusion and sentiment analysis. Multimodal fusion is concerned with reconciling data from more than one modality, to provide a full insight into an emotional experience. As stated by Pei et al., “[m]ultimodal emotion analysis considers the complementarity of emotion expression among modalities and is thus more robust and aligned with natural human behavior expression” (2024, pp. 4). This aspect of AffC aims to make a computer’s affective

abilities as close to the human experience, which consists of simultaneous speech content, vocal intonation, facial expression, gesture, posture, and physiological reactions. Sentiment analysis, the last of the listed AffC tasks, is “the process of gathering and analyzing people’s opinions, thoughts, and impressions regarding various topics, products, subjects, and services” (Wankhade et al., 2022, pp. 5731). Simply put, it is the task of identifying the emotion or emotional valence of an utterance. Although sentiment analysis can be done manually, the increased accessibility to online databases has made it a powerful tool in machine learning; artificial neural networks (ANN) can be trained on data, to later successfully classify the emotions represented not only in text, but also in multimodal materials (Baberwal et al., 2025, pp. 1). Sentiment Analysis has become a key task of AffC, which can help identify the reception of a product or service by customers and users. More specifically, it is utilised in business analysis, healthcare, marketing, social media monitoring, and even stock market prediction (Wankhade et al., 2022, pp. 5763–5766). The task of Sentiment Analysis is of significant relevance to this thesis, as identifying sadness and anger in text is a part of depression symptom recognition.

3. Mental Health

The notion of mental health has permeated the public domain. As defined by the World Health Organization, “[m]ental health is a state of mental well-being that enables people to cope with the stresses of life, realize their abilities, learn well and work well, and contribute to their community” (www.who.int/health-topics/mental-health#tab=tab_1). With interest rates doubling worldwide from 2021 to now (Google Trends graph for the phrase “mental health”: DOA 11.11.2025), it is to no surprise that the topic has also been of utmost concern to the scientific world, especially in post-pandemic times. This section aims to overview the evolution of attitudes towards mental health conditions, the state of mental health research, as well as the options available to individuals battling with mental health-related issues.

3.1. Attitudes towards mental health through time

The perception of health issues related to the mind has changed radically over time. What was at first a topic seen as a taboo in the Middle Ages (where mental health conditions were described as “insanity”, and the affected were hidden away in family homes), proceeded to become the main subject of a medical boom in the mid-20th century stemming from the popularisation of psychiatry and psychology (Abma in Jansz & van Drunen, 2004).

Nowadays, it may seem impossible to imagine a world where the topic of mental health does not exist in the public eye. According to data collected in 2021 by the World Health Organisation (WHO), over 1 billion individuals worldwide suffer from some form of mental disorders, the most common of which are depression and anxiety (2025). In the United States, “[f]rom 2018 to 2021, an annual average of 1,026,000 visits were made by children and adolescents with a diagnosis of a mental health disorder” (Santo et al., 2023). This alone highlights the scale of the problem. Naturally, access to information on the topic has also increased; content concerning mental health is readily available, and can be found in scientific journals, medical pamphlets, self-help books and magazines, as well as social media. However, this was not always the case.

While mental illnesses have accompanied the human species throughout our evolution, they have not always been considered illnesses or disorders. The early interpretations of the causes of mental illness were predominantly linked to supernatural forces, especially demonic possession or divine wrath. Such beliefs proliferated among the Ancient Hebrews, Ancient Greeks, Ancient Arabs, and Christians of the Middle Ages (Horosan et al., 2024). This perception of “madness” gave rise to rampant stigmatisation of the mentally ill. Later, the Renaissance period introduced a significant interest in mental health, especially in the domain of the arts (Szwach and Wojciechowska, 2024). It was also at this time that the first custodial asylums began to open around Europe, with the first being founded in Valencia in 1409 (Abma in Jansz & van Drunen, 2004). However, with these asylums being under the control of the Catholic Church, the mentally ill were still outcast from society and isolated. According to Horosan et al. (2024, pp. 116), “[t]reatment was an eclectic mix of magic, religion, supernaturalism, and medicine, which suited both the educated population and common people, all believing that healing diseases required the reconciliation of these methods”. While interest in mental health was rising, the aim was to entertain the masses by portraying mental illnesses in art and literature, instead of humanising the mentally ill, and creating a place where they could function as a part of society.

As did many other scientific endeavors, the search for the explanation of mental disorders gained traction during the Enlightenment. It was at this time that scientists began to associate “madness” with possible health issues. The dawn of psychiatry can partially be attributed to William Battie’s *Treatise on Madness*, which was published in 1758. In it, Battie characterises “madness”, and says:

“[...] we may collect that Madness with respect to its cause is distinguishable into two species. The first is solely owing to an internal disorder of the nervous substance: the second is likewise owing to the same nervous substance being indeed in like manner disordered, but disordered *ab extra*; and therefore is chiefly to be attributed to some remote and accidental cause.” (1758, pp. 43–44)

Battie not only provided critique for the contemporary state of mental health treatment (or lack thereof), but attempted to characterise “Madness”, proposing that it could either be caused by internal factors, or external triggers. Following these beginnings, medical scholars began to debate organic and non-organic causes of mental disorders, the former of which were decisively prevalent (Abma in Jansz & van Drunen, 2004). Organic causes described mental illnesses as biological problems within the body, whereas non-organic causes believed that the source of the problems could be found in the mind. Supporters of the first school searched for somatic explanations of several maladies with noticeable physical symptoms, leading to the popularisation of misguided theories, e.g. one which “attributed hysterical symptoms to imbalances in a universal magnetic fluid found in individuals” (Farreras in Biswas-Diener & Diener, 2021). However, no convincing methods of combatting the presumed biological causes of mental disorders could be found.

The lack of treatment options linked to the organic-cause theory gave rise to the moralist movement, rooted not in medicine, but in social principles. The moral approach “actively sought to transform the lunatic, to remodel him into something approximating the bourgeois ideal of the rational individual” (Scull, 1975, pp. 227). The main aim of the treatment method was to impose authority over the mental ill, in order to give them a sense of direction. Patients remained isolated in asylums and retreats; it was believed that providing them with a tranquil environment would have a positive impact on their mental well-being. However, due to the lack of agreement between the social and biological schools of mental health research, “the golden age of psychiatry had ended in disillusionment” (Abma in Jansz & van Drunen, 2004, pp. 4). Despite the great increase in interest surrounding the field, little progress had been made.

The post-Enlightenment period of stagnation was thawed by the introduction of nervous disorders, proposed by George M. Beard in *American Nervousness* (1881). Beard states that:

“By the term nervous diathesis we design to express a constitutional tendency to diseases of the nervous system. It includes those temperaments, commonly designated as nervous, in whom there exists a predisposition to neuralgia, dyspepsia, chorea, sick-headache, functional paralysis, hysteria., hypochondriasis, insanity, or other of the many symptoms of disease of the central or peripheral nervous system.” (1881, pp. 25)

This theory strayed from focusing solely on “madness”, but instead brought attention to a variety of deviations from the healthy norm. An increasing amount of attention was being paid to the nervous system. The scene of mental health studies was becoming dense with a variety of specialists, all claiming to have solutions: neurologists began to look for treatments for nervous disorders, advocating for private treatment; psychiatrists searched for a way to depart from asylums and bridge the treatment of madness with the treatment of nervousness; clerics were establishing the first attempts at psychotherapeutical approaches, while still putting pressure on the religious aspect of healing (Abma in Jansz & van Drunen, 2004, pp. 6).

It was at this time that Sigmund Freud, a neurologist, introduced his ideas about the causes of mental issues, and proposed a solution to the problem: psychoanalysis. Freud believed that “repressed childhood sexual memories and fantasies caused neurosis when reactivated in adult life”, and that “[s]uch neurosis could be cured by an elaborate technique emphasizing dream analysis, free association, and the working through of a ‘transference-neurosis’” (Shorter, 1997, pp. 184). Both linking behaviours in adult life to childhood experiences, and highlighting the importance of the unconscious influencing the conscious, were novel perspectives. The theory proposed an explanation for the development of mental disorders; moreover, it provided psychiatrists with a toolset to help patients. Freud’s contribution to psychiatry was the bridging of the organic and non-organic approaches: “the same symptoms could be produced by mental stress and by organic lesions, as both were apt to lower the resistance to the break-through of imperfectly controlled unconscious impulses” (Stengel, 1956). Yet, Freud’s suggestions were not accepted enthusiastically by everyone. The field of psychiatry, which had previously been rooted in biology, was now forced to re-orient its focus to psychology; however, ultimately, this did not impede the spread of psychoanalysis among psychiatrists, due to its popularity among the public, and the financial prospects it brought to private psychiatric practices (Shorter, 1997, pp. 194).

In the early 20th century, as psychoanalysis was on the rise, another event upturned the world of psychiatry: the outbreak of World War I. The Great War provided psychiatrists with an incredulous surge of patients; it is known that “[b]y the end of the first year of the war, 111,772 soldiers, exhibiting the entire range of hysterical symptoms known at that time, had passed through military and field hospitals” (Kaufmann, 1999, pp. 132). Psychiatrists and psychoanalysts were needed more than ever, and they marked their presence in the trenches and in field hospitals, implementing techniques from all contemporary branches of

psychiatric theory. Early attempts at diagnoses were closely linked to organic psychiatry; among others, the soldiers were diagnosed with a condition then called “shell shock”, which was treated by sending soldiers away from the trenches to rest and recuperate (Myers, 1915; Shephard, 2001). Organic psychiatrists believed it to be a physical reaction caused by exposure to explosions. Presently, the condition is formally known as Post-Traumatic Stress Disorder (PTSD). As early as 1915, psychoanalytic methods, such as hypnosis, were also being tested on the battlefield, with professionals explaining neurotic conditions to be unconscious attempts at self-preservation (Brunner, 1991, pp. 353). A third branch of psychiatrists introduced “surprising treatment”, which consisted of: administering electric shocks, isolating soldiers, conducting fake surgeries to scare soldiers, and other treatments which were to the liking of the individual psychiatrist (Brunner; 1991, pp. 355).

The war provided psychiatrists with freedom to test on a large group of patients by means of trial and error, while facing very little consequence. The period has been described as “an experiment of near-gigantic dimensions for psychiatry” (Kaufmann, 1999, pp. 134). As stated by Brunner:

“By creating a climate in which doctors could exercise unlimited power in the name of science and the state, army hierarchy and conditions of war had generated an opportunity for medical violence. It appears that war doctors were allowed to violate the bodies and minds of the soldiers in a way which about a century earlier had been declared illegal for modern criminal justice.” (1991, pp. 355)

While providing more patients than ever before, the war also exerted a large amount of pressure on the psychiatric field. Many misconducts of war psychiatrists can be explained by the political conflict at hand; specialists were expected to quickly prepare soldiers for their return to the battlefield, even if their maladies were not capable of being easily ‘fixed’ (Shephard, 2001). For these reasons, World War One is difficult to summarise from the psychiatric perspective. On the one hand, this period helped psychiatrists understand the complex nature of PTSD, and leave behind the asylum treatment for a more hands-on approach called “forward psychiatry”; on the other hand, it was also a time of gross misconduct and abuse, and the advances which were made were largely ignored after the war subsided (Jones & Wessely, 2014, pp. 1712).

After the end of the Great War, in contrast to the gunfire, the mental damage which had been dealt did not cease. During the interwar period, and the duration of World War II, psychiatrists continued to experiment with a variety of medical and therapeutic approaches, to aid the growing number of patients. At this stage it was clear that sending people to asylums

was a temporary and ineffective solution, while psychoanalysis was seen as an option for the wealthy, who were interested in self-insight; for these reasons, alternatives were explored (Shorter, 1997, pp. 235). New physical approaches to treating mental illnesses gained traction among psychiatrists, who often opted for unproven methods. Some of the most obscure treatments included purposefully exposing patients to malaria, while others consisted of applying convulsive drugs, inducing epileptic fits, employing Electro Convulsive Therapy (ECT), or conducting lobotomies—procedures during which surgeons would sever a patient's prefrontal cortex from the rest of the brain (Shephard, 2001, pp. 206). Simultaneously, a more social approach was developing. An alternate theory for the cause of mental disease was the imbalance of social relations, which led physician Joshua Bierer to pitch the idea of group psychotherapy. Bierer then continued to open the first day hospital: an alternative to asylums, which patients could visit for group therapy sessions and counseling (Shorter, 1997, pp. 285–289). Closer to the front lines, physicians Roy R. Grinker and John Spiegel were developing their own method of psychotherapy, which required soldiers to re-experience their trauma during sessions. By employing this method as soon after the occurrence of the trauma as possible, Grinker and Spiegel achieved high rates of recovery among soldiers (Pols, 2006). Quoting Sarason, Shorter writes that “American psychiatry before World War II was biological psychiatry and within a few years after the war it was largely a psychoanalytical psychiatry” (Shorter, 1997, pp. 213). While the physical solutions, like lobotomies, led to controversies, psychotherapeutical approaches showed promising results, and psychoanalytical approaches retained their pre-war popularity.

Post World War II, through to the 1970s, clashes continued within the community of specialists dealing with mental disorders. This era saw the second rise of biological psychiatry; studies revealed how certain diseases could potentially be linked to genetic factors, and moreover, the first highly successful drug, chlorpromazine, was introduced. Heinz Lehmann had tested the latter on schizophrenic patients, and found that more than half of them ended the treatment period symptom-free (Shorter, 1997, pp. 308). This was a huge breakthrough for drug treatment in psychiatry, as chlorpromazine did not limit other intellectual functions, in contrast to previously tested medicines. Intuitively, while medication-lead psychiatry was gaining traction, and patients' symptoms were being held under control, deinstitutionalisation policies began to be introduced, with “the census of public psychiatric hospitals decreasing by 70 percent” between the 1950s and 1970s (Geller, 2000). The primary reasons for these policies were: overcrowding and underfunding, the advocacy for a greater focus on social treatment meant to move patients out of the institution,

and back into society, and the antipsychiatry movement. Antipsychiatry could be seen as the public's retaliation to the uncontrolled proceedings of specialists during the World Wars; social psychiatrist Thomas Szasz was quoted to say that "what the rack and the stake were to the inquisition, what the concentration camp and the gas chamber were to National Socialism, the mental hospital and electroshock are to institutional psychiatry" (Shorter & Healy, 2007, pp. 187). Antipsychiatry was also present in pop culture at the time, popularised by films such as *One Flew Over the Cuckoo's Nest* (1975), which portrayed mental hospitals as institutions riddled with gross misconduct towards patients. While the movement may have attempted to demonise psychiatry, its effects on the field were positive. By being scrutinised so carefully, psychiatrists were forced to refine classifications of mental disorders, and to be more cautious when diagnosing patients (Berlim et al., 2003).

On the cusp of the 20th and 21st centuries, psychiatry remained a controversial field, and it continues to be one now. The need for standardised classification has resulted in professionals inventing a unified framework for facilitating the diagnosis of mental disorders; these classifications are found in manuals, namely the Diagnostic and Statistical Manual of Mental Disorders (DSM) released by the American Psychiatric Association (APA), and the International Classification of Diseases (ICD), maintained by the World Health Organisation (WHO). The DSM is currently available in its fifth edition (APA, 2013), and the ICD-11 was released in 2022. The manuals provide criteria necessary for segregating symptoms and diagnosing patients, which has given psychiatry the structure it may have lacked in the past. Conversely, some voices within the field believe that the DSM and ICD should be approached critically; Summerfield (2008) points to the fact that the manuals are primarily Western creations, which may be applicable in Western societies, but in turn fail to express cultural inclusivity for individuals from other regions of the world. Other critiques concern the fact that the symptoms listed in diagnostic manuals are linked to social norms, and continue to leave room for subjective judgements made by physicians (Grech & Grech, 2021, pp. 64). Controversies are not only present at the diagnostic stage, but also appear in discourse surrounding the treatment of mental disorders. The treatment of disorders has advanced owing to "the development of new drugs, neuroscience, molecular genetics and Cognitive Behavioural Therapy (CBT)" (Grech & Grech, 2021, pp. 67). However, some critics highlight the importance of remembering that the knowledge specialists possess today remains rooted in hypotheses, and not facts. This has led to the emergence of a postpsychiatric movement, which, in contrast to antipsychiatry, does not aim to demonise the field, but instead

underscores the importance of social, cultural, and ethical aspects, empowering the patient, and not only the specialist (Bracken & Thomas, 2001).

In summary, since its dawn, mental health has been an ephemeral topic. In an attempt to tackle the subject, the field of psychiatry emerged and has ever since been torn between many schools and theories. This is true both in terms of beliefs surrounding the roots of disorders (moral weakness, physiological issues in the body, physiological issues in the brain, nerves, genes, sexual fantasies from early childhood, trauma, natural reactions to the environment, and more), as well as in proposed treatment approaches (asylum care, psychosurgery, medication, Electro Convulsive Therapy, psychoanalysis, group psychotherapy, private therapy, and more). As attitudes towards mental health evolve, psychiatry today is facing a paradigm shift, and must undergo changes, so as to better understand the problem at hand holistically; “[t]he traditional ‘one-size-fits-all’ approach to diagnosis and treatment should be reconsidered and new diagnostic and therapeutic tools should be offered” (Jakovljevic, 2013, pp. 205). The direction the field will take in the future is uncertain, and the foundations of psychiatry and psychotherapy continue to be widely debated.

3.2. Contemporary Mental Health: State, Direction, and Relevance

In current times, an individual searching for help with mental illness has a wide range of services to choose from. According to recent data, in Europe alone there are approximately 90,000 psychiatrists (Eurostat, 2018) and 120,000 psychotherapists (European Association for Psychotherapy, 2025). In the United States, these numbers equal approximately 60,000 (Kaiser Family Foundation, 2024) and 200,000 (US Bureau of Labor Statistics, 2026), respectively. Within psychotherapy alone, there exist a plethora of different methods and approaches (Analytical, Client-Centered, Rational-Emotive, Behavior, Cognitive, Gestalt, etc.), which vary in form and purpose (Corsini & Wedding, 2010). Apart from visiting professionals in person, there has been a rise in available online services, which also provide a range of mental health aid, like teletherapy, online counselling, 24/7 helplines, chatrooms, guidebooks and manuals (Kauer et al., 2014). Departing from traditional clinical approaches, another trend receiving positive reception within the mental health discourse is mindfulness, an idea derived from Buddhist practices, which advocates for insight and focus on present psychological experiences (Gordon et al., 2015). This may perhaps be linked to the growing trend of “self-help” in the mental health context: more and more individuals decide against

accessing professional intervention, in turn relying on self-help books, movies, and support groups (Norcross, 2000).

Despite seemingly open access to mental health services, many people choose not to seek help, or cannot seek help. Studies find that among the most susceptible social group, adolescents, over 75% of people do not receive access to mental health services (Children's Commissioners, 2016). In a systematic review, Aguirre Velasco et al. (2025) found that the most prominent barrier preventing young people from seeking help was stigma, in terms of how both patients and practitioners were perceived. Furthermore, structural barriers, such as the lack of access to means of transportation, long waiting lists, and high costs, were listed as other impediments (Aguirre Velasco et al., 2025). For these reasons, educating society about mental health remains a crucial challenge for the future.

Growing awareness of mental health-related issues has been of the utmost importance nowadays. Following the self-isolation caused by the outbreak of the COVID-19 pandemic in 2019 and early 2020, the number of people suffering from mental illnesses escalated, with statistics showing drastic increases of even 20 percent in certain areas (Loiwal, 2020). It is too early to assess the long-term consequences of the pandemic on mental health; speculations have included an increase in “loneliness, anxiety, depression, insomnia, harmful alcohol and drug use, self-harm or suicidal behavior [...], cases of domestic violence” and “neurotic disorders” (Kumar & Nayar, 2020, pp. 1–2). In response, the WHO has advocated for the strengthening of universal health coverage, which includes changes in: the delivery of mental health services, the workforce providing the services, the financing of mental health research and treatment, government policies, and the technical action taken to treat and prevent mental illnesses. The WHO also highlights the importance of digitalization of services, as the internet has become a powerful tool in educating and aiding people in need of help (WHO, 2022).

Another reason for the worsening of global mental wellbeing may be the growing impact of social media on daily lives. On the one hand, social media has a seemingly positive effect on individuals; it extends the possibilities for communication, provides room for discussing difficult emotions and situations, and in turn allows for the destigmatisation of many issues, including seeking help for mental health struggles (Naslund et al., 2020). On the other hand, studies have shown that excessive social media use may “increase body image dissatisfaction, increase the risk of addiction and cyberbullying involvement, contribute to phubbing behaviors, and negatively affect mood. Excessive use has increased loneliness, fear of missing out, and decreased subjective well-being and life satisfaction” (Zsila & Reyes, 2023, pp. 1). Such prognoses are particularly worrying when matched with data about social

media use; approximately 95% of adolescents use some form of social media, and over a third claim to use it “almost constantly” (Vogels et al., 2022). It is not only adolescents that put their time towards using social media; approximately 5.4 billion individuals worldwide access social media in some form (Statista, 2025). If studies continue to find negative links between social media usage and mental wellbeing, the future may hold an influx of individuals suffering from mental illnesses, such as generalised anxiety disorder, obsessive-compulsive disorder, or depression. For the above-mentioned reasons, the significance of mental health in the 2020s is steadily increasing.

3.3. Affective Computing in Mental Health Research

In face of the bleak landscape of the world’s mental wellbeing, Affective Computing research tools may be of significant benefit. In a narrative review of the field, Schlicher et al. (2025) state that:

While much of the field still relies on direct mappings between (multimodal) cues and emotion states, the increasing integration of context, temporality, and personalized dynamics suggests a transition toward richer, semantically informed affective computing; moving from the perceptual layer (“What you feel”), through the modeling layer (“Why you feel”), toward the intervention layer (“What should be done”). Therefore, affective computing now plays an increasingly central role in the development of digital mental health closed-loop systems, which are cybernetic frameworks that continuously sense mood, update personalized models, and deliver adaptive interventions in real time. (pp. 4)

In practice, what this means is that many of Picard’s (1997) initial goals for Affective Computing have successfully been achieved. For instance, Digital Mental Health Interventions (DMHI), such as mobile apps, wearables, or chatbots, have already been introduced in an attempt to facilitate the mental health crisis. While some may fret that removing the human element from medical contexts will negatively influence the quality of provided care, it is imperative to remember that aids such as DMHIs are only tools used to improve certain issues deeply ingrained in the medical system, e.g. understaffing, or the reliance on patient self-reports. The ideal healthcare setting is neither entirely human, nor entirely machine; it is found at an intersection of the two, with medical professionals and technology cancelling out each other’s shortcomings.

In an overview of the use of DMHIs in behavioural health, Farsadaki and Griffy-Brown (2025) list various AffC solutions which have been successfully implemented. Key insights can be made, even prior to a patient even being diagnosed. Data on sleep quality or heart rate can be passively collected by wearables or smartphones, and later be analysed by

algorithms to identify potential health issues. Such sensors have even found their use in identifying early signs of conditions like depression and Post-Traumatic Stress Disorder (PTSD) (Torous et al., 2021 in Farsadaki and Griffy-Brown, 2025). Going beyond wearables, the use of AI chatbots in medical settings is becoming increasingly popular; an example of this is Woebot, a mental health computer agent which alerts users and provides support if their inputs exhibit signs of depression or anxiety (Fitzpatrick et al., 2017 in Farsadaki and Griffy-Brown, 2025). Mental health professionals have also benefitted from Virtual Reality (VR) technology, which allows patients to face trauma and practice resilience in controlled settings. This is an ingenious approach, as “the virtual environment can be modified in real time to gradually introduce stressors or provide calming stimuli, maximizing both safety and efficacy” (Farsadaki and Griffy-Brown, 2025, pp. 2). An aspect of AffC in mental health which is overlooked is the creation of online, open-access datasets; representative samples of patient interviews allow for progress in tasks, such as automated symptom assessment, or risk modelling. An example of such an affective dataset is the DAIC-WOZ, which has been implemented in this thesis (Gratch et al., 2014) (see Chapter Two, Section 1). These are only the beginnings of research on mental health through the lens of Affective Computing, however, as stated by Schlicher et al. (2025, pp. 1), findings already “suggest that emotionally adaptive systems can strengthen user engagement, simulate empathy, and support more personalized care”.

4. Depression

As reported by the World Health Organization, “[a]pproximately 332 million people in the world have depression” (www.who.int/news-room/fact-sheets/detail/depression). Generally characterised by lowered mood and the loss of pleasure and interest, the disorder is one of WHO’s priorities for Mental Health Action; this is without a doubt linked not only to the prevalence of the disorder, but also to the substantially increased chances of both suicidal ideation and completed suicide (Cai et al., 2021). For these reasons, research on depression detection and prevention is of utmost importance. This sub-section provides an overview of the history of depression and presents the current state of depression research.

4.1. History of Depression as a Mental Illness

In contrast to many other mental illnesses (see Section 3), depression has had a stable position in the medical canon throughout human history, despite being used to classify a variety of different states, moods, and symptoms (Jackson, 1986, pp. 3). Having been recognised as early as the Hippocratic times, the condition was first referred to as *melancholia*, and was closely linked to the humoral theory (Horwitz et al. in DeRubeis, Strunk & Nathan eds., 2017, pp. 11). In fact, the phrase *melancholia* itself literally translates to “black bile”, one of the four humors alongside blood, yellow bile, and phlegm, which were thought to govern human health by the Ancient Greeks (Jackson, 1986, pp. 7). A disturbance in the equilibrium of the humors was thought to cause illness and disease. Therefore, in ancient times, depression was seen as a distinct disease caused by an excess of black bile in the body. Among other post-Hippocratic philosophers was Rufus of Ephesus, who further specified the characteristics of *melancholia*, stating that “those suffering from melancholia were gloomy, sad, and fearful; [...] the chief signs were fear and doubting, with a single delusional idea and yet quite sane on all other matters” (Jackson, 1986, pp. 36). While the humoral theory may long have been disproven in medicine, such statements show that humans have had an understanding of depression since as early as the 1st century AD.

The thinking of the Ancient Greeks and Romans was widely accepted and remained unquestioned for centuries, however, the topic was of no less interest to scientists and philosophers. In 1621, Robert Burton, an Oxford academic, published *The Anatomy of Melancholy*, in which he provided possibly “the most comprehensive description of depression ever compiled” (Horwitz et al. in DeRubeis, Strunk & Nathan eds., 2017, pp. 14). Burton, being part of the classical tradition, attributed the main cause of melancholy to the imbalance of black bile in the body, similarly to his predecessors. The scholar defined the condition as “a kind of dotage without a fever, having for his ordinary companions fear and sadness, without any apparent occasion” (www.gutenberg.org, DOA 02.03.2026). Significantly, Burton highlighted the importance of context in diagnosing melancholy; “apparent occasions” such as grief or sorrow caused by life events were not considered as disordered. When providing insight on how to cure oneself of melancholy, Burton highlighted the importance of seeking help from God, and finding balance in “diet, retention and evacuation, air, exercise of both body and mind, sleeping and waking, and passions or perturbations of the mind” (Jackson, 1986, pp. 97). Despite containing some outdated

explanations of the cause of melancholy, Burton's thorough *Anatomy* provided accurate insight into depression symptoms and their possible alleviations.

Hypotheses reaching outside of the humoral theory of melancholia appeared as late as in the 18th and 19th centuries (Horwitz et al. in DeRubeis, Strunk & Nathan eds., 2017, pp. 13). Leaving behind the theory of the humours, theorists first shifted to the idea of "vapours": "supposed exhalations of the viscera that would rise in the body to affect the brain" (Shorter, 2013, pp. 17). However, this theory was just as unempirical as the four humours. After, the time period saw a similar shift as general mental health research (see Section 3), with theorists splitting roughly into two schools: those who saw *melancholia* as a disease of the "nerves", and those who believed it to be a disease of the brain. It is important to mention, however, that ideas surrounding *melancholia* were plentiful, vast, and unorganised; the term "school" is used as a loose grouping of thoughts appearing at a similar time. The former school largely saw depressed mood as a result of human physiology, and ideas branched out into diagnoses like "neurasthenia", a non-psychotic disorder of the nervous system. Such conditions were treated by general practitioners and neurologists, as they were not seen as disorders of the mind (Horwitz et al. in DeRubeis, Strunk & Nathan eds., 2017, pp. 15). In contrast, the latter school believed that depressed mood was rooted in the brain; thinkers like Esquirol proposed to replace the term *melancholia* with *lypémanie*, attributing it to a partial disorder of the brain, while others, like Griesinger, went even further, establishing the notion that "mental disease is brain disease" (Shorter, 2013, pp. 96). This type of depressed mood was treated by psychiatrists and alienists, often in an asylum setting (Horwitz et al. in DeRubeis, Strunk & Nathan eds., 2017, pp. 15). This rift in nosology was the reason for melancholia splitting into numerous smaller disorders and conditions, muddying the classification categories for what is now known as depression.

It was Emil Kraepelin, a German physician, who finally helped remove the humour-related term *melancholia*, replacing it with the word we use contemporarily: depression (Shorter, 2013, pp. 99). As described by Jackson, Kraepelin:

"began the process of organizing his two, soon to be famous, nosological groups by bringing together in one category psychoses that he considered to be deteriorating in nature and in another category psychoses that he considered to be nondeteriorating. In the first group (die Verblodungsprocesse) were those conditions that were to constitute his notion of dementia praecox in his sixth edition in 1899. In the second group (das periodische Irresein) he brought together mania, melancholia, and circular insanity, thus discontinuing the presentation of mania and melancholia as separate disorders. Then in the sixth edition he replaced the term periodic psychosis with manic-depressive psychosis (das manisch-depressive Irresein) to refer to a group that included manic states, depressed states, and mixed states." (1986, pp. 189)

The changes introduced by Kraepelin achieved two things: the separation of depressive psychoses from what is now known as schizophrenia (called *dementia praecox* by Kraepelin), and the grouping of various mood dysregulations and disorders into the broader category of Manic-Depressive Insanity (MDI). Shorter (2013, pp. 102, 171) approaches these changes critically, claiming that Kraepelin's influence removed *melancholia*'s specificity, turning the nuanced, "age-old" diagnosis into a word which can be applied to a substantial percentage of society. Nonetheless, the shift was permanent, with Kraepelin's terminology and nosology being a major influence on the proceeding classifications of melancholic disorders.

Despite some unifications of definitions appearing, a great confusion continued to exist regarding depression. As stated by Kendell, "there is no consensus of opinion about how depressions should be classified, or any body of agreed findings capable of providing the framework of a consensus" (1976, pp. 25). This assessment of the situation appeared at a time when the American Psychiatric Association had already published its two first diagnostic manuals (see Section 4.2), the DSM-I and the DSM-II. It was the next of these manuals, the DSM-III, which attempted to rewrite the previous two centuries of medical discussions in order to provide a clear outlook on the disease known as depression. Notably, it departed from "diagnostic conceptions about depression over the previous 250 years, and unified depressive conditions into a single category" (Horwitz et al. in DeRubeis, Strunk & Nathan eds., 2017, pp. 19). Furthermore, the DSM-III provided a list of symptoms and conditions required for establishing a diagnosis. Thus, the process of diagnosing depression became much less context-sensitive, with practitioners now focusing on the symptoms presented by patients, and not on the reasons for the disorder appearing in the first place. The manual redirected the focus of scientific discussion, which has remained as such to present day. Similarly to Kraepelin, it has continued to receive criticism for broadening the definition of the condition to such a state, that it has become applicable to a vast proportion of outpatients (Horwitz et al. in DeRubeis, Strunk & Nathan eds., 2017, pp. 21).

4.2. Diagnosing Depression At Present

The current section provides diagnosis criteria for depression from the two prevalent sources of global diagnostic health information: the *Diagnostic and Statistical Manual of Mental Disorders, 5th Edition* (DSM-5), and the *International Classification of Diseases 11th Revision* (ICD-11). The DSM-5, prepared by the American Psychiatric Association (2013), is

a manual which aims to “summarize characteristic syndromes of signs and symptoms that point to an underlying disorder with a characteristic developmental history, biological and environmental risk factors, neuropsychological and physiological correlates, and a typical clinical course” (DSM-5, 2013, pp. 3). The manual contains classification criteria and details for over 70 disorders, and is an influential source of psychiatric information. The ICD-11, introduced by the World Health Organization (2022), is “the international standard for systematic recording, reporting, analysis, interpretation and comparison of mortality and morbidity data”, and it provides “access to 17 000 diagnostic categories, with over 100 000 medical diagnostic index terms” (www.who.int, DOA 04.02.2026). The manual has been fully digitalised, and, in contrast to the DSM-5 which focuses solely on mental health, it provides information about diseases from every medical branch.

4.2.1. DSM-5

The DSM-5 does not recognise “depression” as single, standalone illness; instead, the manual presents the umbrella term of “Depressive Disorders”, under which the following illnesses are found: disruptive mood dysregulation disorder, persistent depressive disorder (also known as dysthymia), premenstrual dysphoric disorder, substance/medication-induced depressive disorder, depressive disorder due to another medical condition, other specified depressive disorder, and unspecified depressive disorder (DSM-5, 2013, pp. 155). As can be inferred from the names alone, the disorders in this array are context-sensitive, and take into account the factors surrounding an individual’s life (e.g. gender, age, medication use), and not just the end-state within which the individual finds themselves. The DSM-5 continues to clarify that “[t]he common feature of all these disorders is the presence of sad, empty, or irritable mood, accompanied by somatic and cognitive changes that significantly affect the individual’s capacity to function. What differs among them are issues of duration, timing, or presumed etiology” (DSM-5, 2013, pp. 155). These are the base requirements for being diagnosed with a depressive disorder.

Each of the disorders underneath the depressive disorder umbrella has its own diagnostic criteria, and accompanying diagnostic features. The manual claims that the disorder which serves as a baseline for the others is Major Depressive Disorder (MDD). The diagnostic criteria for MDD consist of five major points (A–E), in which point A is a list of nine (1–9) criteria, of which at least five must be satisfied by the patient. The full list of diagnostic criteria for MDD (DSM-5, 2013, pp. 160–161) presents itself as follows:

A. Five (or more) of the following symptoms have been present during the same 2-week period and represent a change from previous functioning; at least one of the symptoms is either (1) depressed mood or (2) loss of interest or pleasure.

1. Depressed mood most of the day, nearly every day, as indicated by either subjective report (e.g., feels sad, empty, hopeless) or observation made by others (e.g., appears tearful).
2. Markedly diminished interest or pleasure in all, or almost all, activities most of the day, nearly every day (as indicated by either subjective account or observation).
3. Significant weight loss when not dieting or weight gain (e.g., a change of more than 5% of body weight in a month), or decrease or increase in appetite nearly every day. (Note: In children, consider failure to make expected weight gain.)
4. Insomnia or hypersomnia nearly every day.
5. Psychomotor agitation or retardation nearly every day (observable by others, not merely subjective feelings of restlessness or being slowed down).
6. Fatigue or loss of energy nearly every day.
7. Feelings of worthlessness or excessive or inappropriate guilt (which may be delusional) nearly every day (not merely self-reproach or guilt about being sick).
8. Diminished ability to think or concentrate, or indecisiveness, nearly every day (either by subjective account or as observed by others).
9. Recurrent thoughts of death (not just fear of dying), recurrent suicidal ideation without a specific plan, or a suicide attempt or a specific plan for committing suicide.

B. The symptoms cause clinically significant distress or impairment in social, occupational, or other important areas of functioning.

C. The episode is not attributable to the physiological effects of a substance or to another medical condition

.

D. The occurrence of the major depressive episode is not better explained by schizoaffective disorder, schizophrenia, schizophreniform disorder, delusional disorder, or other specified and unspecified schizophrenia spectrum and other psychotic disorders.

E. There has never been a manic episode or a hypomanic episode.

Two important notes are added to the abovementioned criteria. The first highlights that the symptoms which can be attributed to a different medical condition should not be taken into account during the diagnostic process; the second recommends that practitioners should “carefully consider” whether the MDD is not a natural response to significant loss (i.e. a

result of the grieving process) (DSM-5, 2013, pp. 160–161). The manual further suggests that the distinction between grief and MDD can be made on the basis of persistence and intensity; grief is described as occurring in waves which decrease over time, while MDD is said to be more persistent, and not linked to specific memories or thoughts (pp. 161). These suggestions further highlight the DSM-5's context-sensitivity. The manual also provides information about the degree of the disorder, and functional consequences of MDD and other related disorders. An individual can be classified as suffering from mild, moderate, or severe MDD. The consequences of suffering from MDD may include high mortality rates linked to suicide, the lack of ability to “attend to basic self-care needs”, becoming mute or catatonic, experiencing more pain and physical illness, and experiencing decreases in social and role functioning (pp. 164–167). Like many disorders in contemporary times, depression is not seen as a binary diagnosis; instead, it is measured on a scale, and, in turn, individuals suffering from depression may not always be easily distinguished from typically functioning individuals.

4.2.1.1. The DSM-5 and Gender-Related Aspects of Depression

The DSM-5 mentions the links between gender and depression in two aspects: prevalence, and diagnostic issues. In terms of depression prevalence, the manual states that “[f]emales experience 1.5- to 3-fold higher rates than males beginning in early adolescence” (pp. 165). This notable difference does not, however, extend to the experienced symptoms, the course of the disorder, the response to treatment, or the functional consequences, according to the DSM-5 (pp. 167). Moreover, it is noted that in women the risk of attempting suicide is greater, however, the risk of suicide completion is lower (pp. 167). While the mention of gender in the DSM-5 is explicit, the manual does not provide any theories about why women are at a greater risk of experiencing depression than their male counterparts.

4.2.2. ICD-11

Similarly to the DSM-5, the ICD-11 recognises an array of depressive disorders, namely: single episode depressive disorder, recurrent depressive disorder, dysthymic disorder, mixed depressive and anxiety disorder, premenstrual dysphoric disorder, as well as other specified and unspecified depressive disorders. According to the manual, depressive disorders “are characterised by depressive mood (e.g., sad, irritable, empty) or loss of pleasure accompanied

by other cognitive, behavioural, or neurovegetative symptoms that significantly affect the individual's ability to function", and should not be diagnosed if any of the symptoms point towards bipolar disorder (icd.who.int/browse/2025-01/mms/en#1563440232, DOA 04.02.2026).

As the ICD-11 does not mention Major Depressive Disorder, and does not highlight a "classic" course of the disorder, the diagnostic requirements for Recurrent Depressive Disorder (RDD) will be presented. The ICD-11 has a different structure to the DSM-5, in the sense that it possesses fewer diagnostic criteria, but instead provides a wide variety of "modifiers", which can be added to further explicate an individual's condition. The base description of RDD is:

"Recurrent depressive disorder is characterised by a history of at least two depressive episodes separated by at least several months without significant mood disturbance. A depressive episode is characterised by a period of depressed mood or diminished interest in activities occurring most of the day, nearly every day during a period lasting at least two weeks accompanied by other symptoms such as difficulty concentrating, feelings of worthlessness or excessive or inappropriate guilt, hopelessness, recurrent thoughts of death or suicide, changes in appetite or sleep, psychomotor agitation or retardation, and reduced energy or fatigue. There have never been any prior manic, hypomanic, or mixed episodes, which would indicate the presence of a Bipolar disorder." (icd.who.int/browse/2025-01/mms/en#1194756772; DOA 06.02.2026)

Similarly to the DSM-5, ICD-11 points towards accompanying symptoms of depressive disorders, such as feeling unnaturally tired and slowed down, having issues with sleeping and eating, as well as lacking the ability to concentrate. The specifiers proposed by the ICD-11 diagnostic system are:

- 6A71.0 Recurrent Depressive Disorder, Current Episode Mild
- 6A71.1 Recurrent Depressive Disorder, Current Episode Moderate, without psychotic symptoms
- 6A71.2 Recurrent Depressive Disorder, Current Episode Moderate, with psychotic symptoms
- 6A71.3 Recurrent Depressive Disorder, Current Episode Severe, without psychotic symptoms
- 6A71.4 Recurrent Depressive Disorder, Current Episode Severe, with psychotic symptoms
- 6A71.5 Recurrent Depressive Disorder, Current Episode, Unspecified Severity
- 6A71.6 Recurrent Depressive Disorder, currently in partial remission
- 6A71.7 Recurrent Depressive Disorder, currently in full remission

(icd.who.int/browse/2025-01/mms/en#1194756772; DOA 06.02.2026).

These specifiers allow for practitioners to assess the severity of a depressive episode, as well as to provide information about whether or not the patient is also experiencing psychotic symptoms. For RDD, a specialist may also specify whether the patient is in partial or full

remission, meaning that they are recovering from the disorder. Additionally, the diagnosis can also be modified to include symptoms which may occur alongside RDD. These modifiers are: with prominent anxiety symptoms (6A80.0); with panic attacks (6A80.1); current Depressive Episode persistent (6A80.2); current Depressive Episode with melancholia (6A80.3); with seasonal pattern of mood episode onset (6A80.4); mental or behavioural disorders associated with pregnancy, childbirth or the puerperium, without psychotic symptoms (6E20); mental or behavioural disorders associated with pregnancy, childbirth or the puerperium, with psychotic symptoms (6E21). As can be seen from this complex, multi-step diagnostic process, the ICD-11 provides complex and thorough means of determining a patient's current condition, taking into consideration both their medical history, as well as all their co-occurring symptoms.

4.2.2.1. The ICD-11 and Gender-Related Aspects of Depression

In terms of correlations between gender and depression diagnoses, the ICD-11 provides similar information to the DSM-5. Accordingly, the ICD-11 states that the risk of suffering from depression is twice as high in women, and that as a gender group they are more likely to attempt suicide. However, the manual further details that despite lower attempt rates, men are more likely to die by suicide. A novel addition is that “[w]omen with a diagnosis of a Depressive Disorder are more likely to experience co-occurring Anxiety or Fear-Related Disorders, disturbances in appetite, and weight gain whereas it is more common for men to experience co-occurring alcohol and other Disorders Due to Substance Use, poor impulse control, and increased risk-taking behaviour” (icd.who.int/browse/2025-01/mms/en#1194756772; DOA 06.02.2026). This statement points towards a notable discrepancy, as the DSM-5 manual claimed that no differences in symptoms occurred between men and women.

4.3. The State of Depression Research and Treatment Now and in the Near Future

Despite the seeming agreement on how depression should be diagnosed, studies on the disease continue to be of significant importance; contemporarily, researchers task themselves with analysing the cause of the disease, expanding the range of treatment possibilities, and working on methods of depression prevention and detection. These tasks have become of increased importance since the outbreak of the COVID-19 pandemic; in one study,

researchers “estimated a significant increase in the prevalence of both major depressive disorder (with an estimated additional 53·2 million [95% uncertainty interval 44·8–62·9] cases worldwide—ie, a 27·6% [25·1–30·3] increase) and anxiety disorders (76·2 million [64·3–90·6] additional cases—ie, a 25·6% [23·2–28·0] increase) since before the pandemic” (Taquet et al., 2021). As the percentage of depressed individuals is on the rise, it is paramount for people, both academics and laymen, to pay particular attention to the disease, finding new solutions and bringing awareness to the topic.

In terms of identifying the causes of depression, the scientific world has largely turned to inspecting the process from the neurochemical and neurobiological perspectives. Importantly, a plethora of studies have found links between neuroinflammation and MDD (Pearson et al., 2024). A neurotransmitter thought to be of key importance is serotonin; formally known as 5-hydroxytryptamine, the substance is responsible for modulating almost all behavioural processes in humans, from mood regulation to motor control (Berger et al., 2009). In fact, early research claimed that serotonergic dysfunction was responsible for most types of depression (Holtzheimer & Nemeroff, 2006), and these findings have permeated the human consciousness, becoming somewhat of a household fact. Such ideas have taken hold largely due to “the effectiveness of selective serotonin reuptake inhibitors (SSRIs) and selective serotonin and norepinephrine reuptake inhibitors (SNRIs) in treating depression, which are effective in 70% of MDD patients” (Pearson et al., 2024, pp. 69). SSRI and SNRI medication is responsible for slowing down the process of serotonin reabsorption by neurons in the brain, therefore prolonging the neurotransmitter's effects. Prescribing this kind of medication has become common practice for depression treatment. However, recent research shows that while drugs which target serotonin prove to be effective, serotonin itself may not actually play a key part in causing depression; some researchers claim that “no evidence supports depression stemming from decreased activity or concentration of serotonin”, and instead point to vaster internal processes occurring in the kynurenine pathway, where tryptophan, an essential amino acid, is metabolised (Pearson et al., 2024, pp. 70). There is still much to discover and confirm in the field of biology-oriented depression studies. This is of key importance, as a better understanding of the behaviour of neurons and neurotransmitters in the Central Nervous System (CNS), and the human body as a whole, can create new possibilities for matching the correct medication to a patient's symptoms.

Another goal for depression-related research is to further the development and popularisation of psychosocial forms of treatment, like psychotherapy. This form of treatment provides individuals with a long-term and sustainable form of managing the illness, as

“[p]sychotherapy for patients with depression strengthens the therapeutic alliance and enables the patient to monitor their mood, improve their functioning, understand their symptoms better, and master the practical tools they need to cope with stressful events” (Karrouri et al., 2021). It is often recommended alongside pharmacotherapy, and seen as equally important to the remission process (Dunlop, 2016). There is no “one-size fits all” variant of therapy, and different types are suggested to people struggling with depression, depending on their symptoms, the disorder’s intensity, comorbidities, and remission state. Among therapies targeted towards depressed individuals, one can find Cognitive-behavioural Therapy (CBT), Interpersonal Therapy (IPT), Supportive Therapy (ST), and others. Current research on depression is taking a network-oriented approach, looking at the interplay of many biological and social interactions; taking this into consideration, Barchas & Brody (2015, pp. 9) make a prognosis about the state of depression research in the future, pondering whether:

Will we enter an era in which tests for types of depression, including severity and suitable treatment, will be based on studies of multiple fields and sources, such as behavior, feelings, genetics, hormones, neuroregulators, brain anatomy, and diagnostic imaging? It is clear that behavior both relies on and influences neurobiology, and that changes in neurobiology can have both short- and long-term effects on behavior—the relationships are reciprocal. Thus, advances in understanding both behavior and the components of neurobiology require openness to inquiry in both areas.

It is possible that future research will simultaneously take this multi-factored, holistic approach, with the aim of bettering science's understanding of depression. By acknowledging the bidirectional connection between neurobiology and social behaviour, scientists could develop treatment methods with a greater precision, distinguishing between various subtypes of depression (Barchas & Brody, 2015).

Apart from biologically- and socially-oriented therapies, current research is also focusing on a third branch of solutions, known as somatic treatments. Among these, one can find: electroconvulsive therapy (ECT), transcranial magnetic stimulation (rTMS), transcranial direct current stimulation (tDCS), vagus nerve stimulation (VNS), deep brain stimulation (DBS), and magnetic seizure therapy (Karrouri et al., 2021). Such treatment requires the use of devices which stimulate given brain regions from the outside in. Currently, somatic methods are often recommended as a “last resort” to individuals who suffer from treatment-resistant depression (TRD), as pharmacotherapy and psychotherapy do not suffice in their case. Of these somatic treatments, ECT is the only one with confirmed results for certain depression types (Karrouri et al., 2021). This may result from the fact that it has been the most thoroughly researched. However, some sources suggest that “rTMS is safe and well

tolerated, and it has been approved by the FDA for adults with depression who have failed to respond to one antidepressant” (Cusin & Dougherty, 2012, pp. 7). Relative lack of data on the remaining somatic treatment options is a challenge for depression research to tackle in the future.

Depression prevention is of equal importance to depression treatment. For this reason, researchers are also working towards better methods of depression detection, a field which has benefitted from the rise of Natural Language Processing (see Section 1), Affective Computing (see Section 2), Artificial Intelligence and Machine Learning. If the diagnostic process was to be facilitated, individuals in need of help would have quicker access to better-suited treatment, preventing the further development of symptoms. The traditional process of diagnosing depression is based on subjective judgements of both the patient and the practitioner; patients are expected to describe symptoms to the best of their ability or to fill in questionnaires, and practitioners are then tasked with discerning which diagnosis is the best suited to what the patient has described. Instead, the interdisciplinary task of depression detection aims to make the process objective and data-driven (Ren et al., 2025, pp. 2). With access to large corpora of multimodal data, which contain representative samples of both depressed and non-depressed individuals, “AI-driven systems can analyze textual, auditory, and biometric inputs to identify subtle behavioral markers of mental distress” (Mikaeili et al., 2025, pp. 7355). This novel approach exhibits two major advantages: firstly, diagnoses are made on the basis of patterns found in quantifiable data, leaving less room for human error, and secondly, such corpora contain a scope of information which is not generally accessible to human practitioners, e.g. biometric data collected by wearables (smartwatches etc.). The intersection of computer science and medical research provides room for further developments in research on depression, and mental health in general.

5. What Language Can Reveal About Mental Health State and Gender

As it has been insinuated throughout this thesis, the way in which an individual communicates can reveal information about their emotional state (see Section 2.2), or, possibly, provide insight on their mental well-being (see Section 4.3). As stated by Tausczik and Pennebaker (2009, pp. 25), “[l]anguage is the most common and reliable way for people to translate their internal thoughts and emotions into a form that others can understand. Words and language, then, are the very stuff of psychology and communication. They are the medium by which cognitive, personality, clinical, and social psychologists attempt to understand human beings”. Simply put, language both encodes and decodes human thought,

providing a “doorway into the mind” (Savekar et al., 2022). Since mental health is closely linked to cognitive behaviours and emotional states, language has also been found to be a useful metric for assessing mental state, or at least identifying symptoms of possible disturbances to an individual’s mental well-being. Among mental disorders, depression has been thoroughly tested in terms of linguistic markers associated with the condition; the links between depression and language have been present in both descriptive works, such as Beck’s cognitive model of depression (1967), as well as many theoretical works (Newell et al., 2018; Zimmerman et al., 2017; Al-Mosaiwi and Johnstone, 2018; see Section 5.2). This has also been tested neuroscientifically, with studies showing that brain areas which are activated during semantic processing, such as the gyrus and prefrontal cortex, behave abnormally in subjects with depression (Abdullaev et al., 2002). As this disorder is the focal point of this thesis, this sub-chapter will focus specifically on the bidirectional links between communication and depression.

5.1. Non-Verbal Aspects of Communication and Depression

It is important to remember that communication is made up of not only what is being said, but also of how something is being said, and what is being done by the interlocutor in the meantime. In other words, communication consists of both verbal content and non-verbal content. It is said that non-verbal aspects of communication, like posture, facial expressions, bodily gestures, eye-contact, tone of voice, and speech rate, are responsible for conveying information about emotions and attitude, making up approximately 90% of effective communication (Park and Park, 2018). Moreover, “[i]ndividuals who are functioning under a stressful or emotional state of mind will often communicate even more useful information in a paraverbal or non-verbal manner” (Hannigan et al., 2020, pp. 224). As depression is linked to a negative emotional state and increased stress, it may be theorised that there are differences in non-verbal communicative behaviour (NVB) between healthy and depressed individuals, and that these aspects of NVB could be helpful in recognising and diagnosing a depressed state.

Research on the possible correlation between bodily movement and depressed mood began as early as the 1960s. Early studies were predominantly observational; despite this limitation, some tendencies in the motor behaviour of depressed individuals were noted. Grinker et al. (1961) reported that depressed patients exhibited a preference to sit alone

quietly, gave slower responses in conversation, and simultaneously appeared to be more restless (Ellgring, 2007, pp. 24). Later, researchers attempted more experimental designs, in order to get better insight into patients' NVB. In a study which analysed the movement patterns of 13 depressed patients at the moment of being admitted to a psychiatric institution, as well as at the moment of their discharge, Fisch et al. (1983) found that during a depressive episode, individuals spent less total time in motion, and displayed less complex patterns of movement. While individual differences played a big role, these early findings suggested that depressed individuals were more inhibited in their display of behaviour, and had issues with achieving successful communicative exchanges (Fisch et al., 1983, pp. 317). A contemporary study examined the valence of non-verbal behaviour (NVB) expressed by healthy and depressed individuals, and its findings suggest that patients diagnosed with Major Depressive Disorder exhibit significantly more negative NVB (Fiquer et al., 2018). This includes an increase in patients shrugging, keeping their heads down, and using more adaptive, self-soothing hand gestures. It is of significance to mention that all such impediments to NVB have a real impact on depressive patients, as "non-verbal communication processes are involved in the risk of recurrence of depression. Remitted people's difficulties in reaching non-verbal convergence with others may hamper them in their interpersonal functioning and, as a consequence, may put them at risk for new episodes of depression" (Geerts et al., 2006, pp. 200). Therefore, bodily NVB is important for both the practitioner and the patient: successfully identifying trends in a patient's NVB facilitates depression detection, and a patient's ability to non-verbally communicate impacts their functioning in society.

Another example of NVB are facial expressions. Research has shown that both facial expression production and recognition are impaired in patients suffering from depression. In terms of the former, Girard et al. (2013, pp. 1) found that depressed patients "made more facial expressions associated with contempt, smiled less, and those smiles that occurred were more likely to be accompanied by facial actions associated with contempt". The researchers believe that this finding is congruent with the theoretical explanations which state that depressed individuals have a tendency to withdraw from society and isolate themselves. With regard to facial expression recognition, there exists a large body of literature stating that people struggling with depression have a general tendency towards misclassifying emotions; researchers theorise that this may be due to a lack of attentiveness towards others which is caused by closing in on oneself (Persad & Policy, 1993). Other studies report that depressed individuals experience a "negative shift", meaning that they specifically misclassify positive emotions as being neutral (Demenescu et al., 2010). This effect may be the result of the

negative mental state of MDD patients. Moreover, research suggests that even the detection of positive emotions may be impaired in depressed individuals; Suslow et al. (2001, pp. 857) found that “[d]epressed subjects were significantly slower in responding to positive faces than normal subjects”. The way in which people with depression conceal facial emotional cues and misinterpret the emotions of others only increases the extent of their social isolation.

As they are vocal, the prosodic features of a given utterance may appear to reach outside of the class of NVB. However, prosody is not inherently linked to the content of speech, but instead to its phonological characteristics; for this reason, acoustic properties of speech are also considered to be non-verbal behaviours, and function under the label of paralanguage (Trager, 1961). Studies have found that a range of paralinguistic traits may also serve as valuable biomarkers of depression. For instance, research shows that patients with depression exhibit abnormally high shimmer and jitter values, where jitter is understood to be the “period-to-period variability of the pitch period in glottal period duration”, and shimmer is the measure of the “micro-perturbations in vocal folds vibration amplitude” (Seo et al., 2024, pp. 7). It is believed that these discriminating differences may be the result of neurophysiological abnormalities which cause muscle tension and affect the individual’s control over their vocal tract. In contrast, speech rate, another paralinguistic feature, has been found to be lowered in MDD patients, as a result of the general psychomotor retardation caused by the disorder (Sobin and Sackheim, 1997). By analysing the correlations between individual phone durations and specific subtopics of the HAMD depression scale (Hamilton, 1960), Trevino et al. (2011) found “distinct phone-specific relationships”; this suggests that given symptoms of depression affect the speech rate of particular sounds differently. Practitioners could potentially use these correlations to diagnose the severity of an individual’s symptoms more effectively. Taking spectral features into consideration, research has also shown that “[s]pectral centroid and entropy are higher in non-depressed speech, indicating greater timbre brightness and spectral richness”, whereas depressed individuals exhibit a flattening pattern of speech (Seo et al., 2024, pp. 9). The analysis of the correlations between paralinguistic features and MDD is an increasingly effective diagnostic tool, as the vocal characteristics of speech are affected by physiological and cognitive processes, both of which are atypical in the presence of a depressive disorder.

5.2. Text-Only Analysis of Depression

Despite the compelling relevance of non-verbal behaviours in the process of meaning creation and transmission, the verbal content of an utterance may also serve as a valuable insight into an individual's state of cognitive wellbeing. At a time when automated chatbots are being trained as diagnostic agents (see Section 3.3), the analysis of text data is crucial; not only does it serve as a complementary indicator to various NVB markers in the verbal channel of communication, it also allows for written and digital channels to be taken into consideration. This opens the doors to a future in which an individual would only have to take part in a written interview online to provide their practitioner relevant diagnostic information, bypassing the need to conduct phone call interviews, or the even more time- and resource-consuming face-to-face interviews. As the linguistic modality of language provides incomplete communicative information when analysed without taking NVB and paralanguage into consideration (Park and Park, 2018), the field leaves much to be studied (See Section 1). For these reasons, the text-only analysis of depressive markers is the focus of this thesis. For the specific markers taken into consideration in this thesis, see Chapter Two, Section 3.1.1.

In an attempt to synthesise the theoretical characteristics of depression with the various empirical findings on the linguistic markers present in the discourse of depressive individuals, Savekar et al. (2022) conducted a systematic literature review. By reviewing 25 full-text articles published between 2010 and 2020, the authors presented an up-to-date picture of the text-only linguistic research of depression. It is important to mention that the authors included a range of research papers, some analysing written discourse, and others spoken discourse. The provided theoretical background focused on the physical, social, and psychological symptoms of depression (Rosario et al., 2014), with emphasis on the “emotional dysregulation and frequent mood-switching” present in people diagnosed with depression (Savekar et al., 2022, pp. 192). Their findings across studies indicated that linguistic markers of depressive symptoms fall into three categories: function words, content words, and other language features (pp. 208–209).

Function words, such as determiners, pronouns, prepositions etc., are words which serve a structural purpose in language, yet do not convey a lexical meaning (Fries, 1952, pp. 106). Savekar and colleagues' (2022, pp. 210) literature review found that among function words, pronouns serve as especially useful linguistic markers of depressive behaviour. The First Person Singular Pronoun (FPSP), in both the subjective (I) and objective (me) form, has been found to be a strong indicator of depression across an array of studies. In the analysis of

written discourse, Tackman et al. 2019 found that “[t]he effect was present for all first-person singular pronouns except the possessive type, in all communication contexts except the impersonal one, and for both females and males with little evidence of gender differences” (pp. 2). It is worth mentioning that the finding’s generalisability across sex categories is of particular significance to this thesis. The prevalence of FPSP was further confirmed by O’Dea et al. (2018), who conducted a longitudinal study consisting of blog post analysis. Their results show a significant correlation between FPSP and suicidal ideation, as well as anxiety scores (pp. 10). A study conducted by Zimmerman et al. (2017) found further discriminating indicators within the sub-category of FPSP, reporting that “first-person singular pronoun use significantly predicted depressive symptoms approximately 8 months later, even after controlling for depressive symptoms at baseline or discharge” and that “this effect was mainly driven by the use of objective and possessive self-references such as ‘me’ or ‘my’” (pp. 384). This finding suggests that the objective and possessive pronouns have even more predictive power than the subjective ‘I’ pronoun in text-only depression detection. The findings on FPSP also apply to spoken discourse analysis, with Trifu et al. (2017) reporting elevated FPSP use in a narrative elicitation interview task. Savekar et al. (2022) interpret the findings on FPSP use to be a result of self-focused attention schemas; by closing themselves off from their surroundings, depressed individuals exhibit self-rumination tendencies. This phenomenon has been labelled to be the self-focused theory of language in depression (Jarrold et al., 2011). Outside of first person pronouns, some studies have also focused on the use of the third person. Notably, Bernard et al. (2016) found that in written texts, individuals with depression expressing negative affect exhibited increased third person singular pronoun (he/she) and third person plural pronoun (they) use. Bernard et al. believe that “perhaps individuals experiencing depression or negative affect engage in a form of “other talk” as a means of self-regulation or distraction from negative feelings” (2016, pp. 7). Such theories suggest that depressed individuals shift focus away from themselves, as a way of distracting themselves from their depressed state. This finding is interesting, as it contrasts the previous conclusions of self-focused FPSP studies; however, the third person pronoun effect has not yet been replicated by other researchers.

Content words carry distinct lexical meanings, serving as complementary opposites of function words (Fries, 1952). The linguistic markers of depression linked to content words are connected to the semantic content of words, the verb tenses, and even the frequencies of certain word categories appearing in an utterance. In terms of semantic content and emotional valence, studies have shown that depressed individuals experience emotional dysregulation,

with their mental distress resulting in increased sadness and anger (Savekar et al., 2022, pp. 209–210). Specifically, Lyons et al. (2018) report that words affectively linked to sadness (e.g. ‘grief’, ‘cry’, ‘sad’) were significantly higher in individuals diagnosed with MDD, than any other mental disorder (schizophrenia, bipolar disorder, etc.). This finding is relevant, as other emotion words were significantly elevated in comparison to the healthy controls, but indiscernible between the different disorders; this may mean that “sad” words can be the key indicator of depression symptoms. In a study on verb tenses, Smirnova et al. (2018) found that depressed individuals exhibited a tendency to use more continuous imperfect and past tense action verbs. This was replicated in the previously mentioned study conducted by Trifu et al. (2017); researchers believe that these findings result from the tendency of depressed individuals to dwell and ruminate on events from their past. The continuous and imperfect tense forms have also been interpreted to be forms of linguistic self-focus, as individuals use self-centered time references (Rosenbach and Renneberg, 2015). Finally, word frequency-wise, depressed individuals have been found to use fewer nouns and more verbs and adverbs in comparison to healthy controls (Leis et al., 2019). Among these findings, the use of negative affect words has been the most well-documented linguistic marker of depression.

The category of “other language features” proposed by Savekar et al. (2022) contained elements such as: style, sentence structure, and cognitive process words (pp. 209). A wide scope study on the two former elements found that people with depression tend to be more narrative and descriptive, rather than analytic, and that they use more colloquialisms and informal words in comparison to the healthy sample (Smirnova et al., 2018). This same study also reported that, syntactically, MDD patients use single-clause sentences more frequently than complex sentences. The latter element of the category results from the idea that depressed individuals may be characterised by a greater mental defeat; the linguistic markers associated with this cognitive state have been found to be the increased use of death-related words (Kleim et al., 2018), and the increased frequency of discrepancies appearing in the narrative of MDD patients (Eischtaedt et al., 2018). What is worth noting with regard to this thesis, among the 25 research articles analysed by Savekar et al. (2022), a single study found a difference in marker use between men and women; Havigerova et al. (2019) report that pronominalisation and action ratio may be valuable markers of depression for men, while punctuation and sentence complexity may be significant for the diagnosis of women. These emerging markers may prove to be useful, however, it is important to note that they come from unreplicated studies.

Upon analysis of the contemporary studies on linguistic markers of depression, Savekar et al. (2022, pp. 213) conclude that “[t]he resultant features of the studies support the idea that the accuracy of depression detection is primarily based on the thematic content analysis of spoken and written discourses of depressive individuals”. The authors find that the lexical content and functional cues of utterances produced by depressed individuals are characterised by tendencies to use negatively-charged language, be self-focused, and to exhibit ruminations on the past. This means that the symptoms of depression observed in a practitioner’s office are often reflected in a patient’s language use. However, the authors highlight the importance of contextual cues that may not be grasped by quantitative analysis methods, like using LIWC (see Chapter Two, Section 3.1), which happened to be the most prevalent methodological tool in this field. Furthermore, Savekar et al. (2022) note that while differences between healthy controls and depressed individuals are statistically significant, finding differences in linguistic markers between mental disorders is a more nuanced task, as they may form a continuum, without clear categorical boundaries. This diagnostic task still poses a scientific challenge. The literature review also reveals that linguistic markers may not be generalisable across languages and cultures (cf. Havigerova et al., 2019; Trifu et al., 2017; Smirnova et al., 2018). As cultural and linguistic differences appear, it may be relevant to assess whether other predictors, such as gender, play a role. This is the primary challenge which this thesis hopes to tackle.

5.3. Language and Gender

The notion that each person has a unique and discernible style of speaking and writing has been accepted and implemented in many scientific fields, such as forensic linguistics and stylometry. Each individual’s idiosyncratic linguistic style is known as an idiolect (Coulthard, 2004). Specialists attempt to find quantifiable linguistic units of style, which can then be used for tasks such as authorship attribution (determining who wrote a text), authorship verification (determining whether a given person wrote a text), or authorship profiling (describing and predicting the traits of an author). Importantly, idiolects can be affected by many factors, both environmental and psychological; these include “personality, mental health, and being a native speaker or not; [...] age, gender, education level, and region of language acquisition” (Daelemans, 2013, pp. 451). As a person’s surroundings and background have a significant impact on linguistic style, it has been suggested that apart from idiolects, “sociolects” can also be established; this implies that “texts by a group of authors,

for instance in terms of gender or time period, share more similarities within a group than between group” (Louwerse 2004). Gender-wise, these idiosyncratic styles have been narrowed down even further, giving rise to the sociolinguistic notion of “genderlect”: the idea that women and men use language in different ways, and for different purposes (Tannen, 1990).

Tannen (1990), one of the theorists behind genderlect, hypothesised the existence of a dichotomy; namely, that women use “rapport-talk”, while men use “report-talk”. Rapport-talk, also called private speaking, describes the fact that women communicate with the purpose of “establishing connections and negotiating relationships” (Tannen, 1990, pp. 111). Conversely, report-talk, also known as public speaking, suggests that men communicate to “preserve independence and negotiate and maintain status in a hierarchical social order” (Tannen, 1990, pp. 112). Tannen (1990) roots these ideas of the public and the private in cultural upbringing. She begins by explaining that women are often raised in a way that discourages them from taking up space in public conversation; this means that they often refrain from participating in discourse, even if they have a strong view on the topic, or if the conversation explicitly concerns them. In turn, they engage in discourse at home, and use this form of contact as a way to build intimacy, seemingly talking more than men. Men face the opposite cultural challenge; they feel a social pressure to have a voice in public, and then treat the environment of the home as a place where they can rest from having to participate in conversation. Tannen suggests that for men, saying a thought out loud instantly grants it merit and importance, as this is what they have been raised to believe and do; therefore, the issue is at heart a contradiction between the purposes of information and intimacy (1990).

From a grammatical standpoint, similar observations were voiced by Lakoff (1973), who noted that societal and cultural norms influence not only how women speak, but also how they are spoken of. She states that:

“It will be found that the overall effect of ‘women’s language’ - meaning both language restricted in use to women and language descriptive of women alone - is this: it submerges a woman’s personal identity, by denying her the means of expressing herself strongly, on the one hand, and encouraging expressions that suggest triviality in subject-matter and uncertainty about it; and, when a woman is being discussed, by treating her as an object - sexual or otherwise - but never a serious person with individual views. Of course, other forms of behavior in this society have the same purpose; but the phenomena seem especially clear linguistically.” (Lakoff, 1973, pp. 48)

Lakoff (1973) believes that from childhood, girls are subject to a form of linguistic oppression, which teaches them what they can and cannot say, as well as how it should or should not be said. This habit, developed in youth, then continues on into adulthood, and

affects women's opportunities in society, making them appear incapable of taking a stance on serious topics. In an analysis of lexis and syntax, the author focused on lexis (specifically colour terms, particles, and evaluative adjectives), as well as syntax (tag-questions and the intonation in regard to making and answering requests). For instance, Lakoff (1973) claims that it is much more socially acceptable for women to make precise colour judgements, such as between "mauve" and "lilac", where men would just say "purple". This is explained by the fact that topics deemed by society as unserious are delegated to women, and that men then "find such discussion amusing because they consider such a question trivial, irrelevant to the real world" (Lakoff, 1973, pp. 49). The same is suggested for particles such as expletives; where a man may exclaim "shit!", a woman could be looked down upon for doing the same. Instead, a safer, more feminine choice may be "oh dear!", or "fudge!". While this may appear to be trivial, Lakoff states that the "choice of particle is a function of how strongly one allows oneself to feel about something, so that the strength of an emotion conveyed in a sentence corresponds to the strength of the particle" (1973, pp. 50). Following this line of thought, it can be said that women are not allowed to express anger or temper as freely as men may be. However, it is also suggested that in terms of adjectives, it is men's speech that may be restricted; while some adjectives (e.g. "cool", "great") are seen as neutral, some are culturally reserved for women (e.g. "charming", "sweet", "lovely"). While a lot has changed over the five decades which have passed since Robin Lakoff published her paper, it is reasonable to believe that using words from the "feminine" class of adjectives may be seen as demasculinizing for some men. In turn, the neutral category is open for all, and freely used by women. The reasoning behind this is explained as so: "[t]he language of the favored group, the group that holds the power, along with its non-linguistic behavior, is generally adopted by the other group, not vice versa" (Lakoff, 1973, pp. 50). It is socially rewarding to speak like the representatives of a group with the higher social class, and socially harmful to speak like the repressed. Syntax-wise, the author finds that women may have a tendency to use more hedging language, and more question tags (e.g. "isn't it?") instead of strong declaratives or direct questions (1973, pp. 56). Such a communication style is less assertive, and, therefore, prevents conflict with the addressee.

The differences in language use between men and women have also been tested empirically, in a quantitative manner. In an analysis of mean sentence length, Mulac & Lundell (1986, 1994) found that women produced longer sentences; however, men produced more words overall. The finding was observed both in spoken (1986) and written (1994) samples. In a comprehensive statistical analysis of a large corpus of texts (approx. 14,000

samples), Newman et al. (2008) found that “[m]en exceeded women on a number of linguistic dimensions including word length, numbers, articles, and prepositions. Men also discussed various current concerns more frequently, and swore more often” (pp. 223). Similar findings were reported in terms of sentence complexity, with a study of written fiction reporting that male authors produced more syntactically complex sentences than their female counterparts (Żyndul, 2025). Newman et al. (2008) further reported that women’s language had a higher probability of containing pronouns, and words linked to sociableness and cognitive processes. Such observations are in line with Tannen’s (1990) and Lakoff’s (1973) theoretical standpoints. With regard to this thesis, a study conducted by Mehl and Pennebaker (2003), may be of particular significance; upon statistical analysis, the authors report that women were more likely to use the first person singular pronoun (FPSP). An increased use of FPSP was also found among individuals with depression (see Section 5.2.), a disorder which affects more women than men (see Sections 4.2.1.1 and 4.2.2.1). With all of the above observations in mind, if men and women display different tendencies in language use, it is possible that these differences will affect other linguistic analyses, such as text-only diagnoses of mental health disorders.

6. Mental Health, Gender, and Sex Through the Lens of Neurofeminism

This thesis aims to focus on differences in language use between men and women; while seemingly simple, such studies are made difficult by the unclear operationalisations of sex and gender, and the lack of widely accepted and widely adopted methodology. White et al. (2021) note that contemporarily, when speaking of humans, sex is popularly accepted to be the biological aspect of “maleness” and “femaleness”, while gender corresponds to the psychosocial aspects of these “categories”. In an analysis of the dilemma in the context of neuroscience, Duchesne (2025) notes that understanding these two dimensions as two independent constructs limits the possibility of analysing their interrelation, which, in her view, is a fatal flaw. Duchesne (2025) approaches the issue from the perspective of neurofeminism, a neuroscientific subfield, which “challenges assumptions regarding the existence, origin, and operationalization of sex differences in the brain and interrogates the validity of translating sex differences observed in nonhuman animal models into explanations for human gender/sex realities” (pp. 116). The author argues for the importance of recognising the biosocial entanglement of sex and gender: sex exceeds biology and is sociopolitically charged, and gender is also a reality of the brain and body, not just the mind. It is not possible to see the two dimensions as completely separate entities; operationalisations

which ignore these facts are insufficient simplifications of inherently complex, intersectional constructs. In line with Duchesne (2025) and Kaiser (2012), when writing about the biosocial intersections of sex and gender, I will use the format of: sex/gender.

While to some, this issue may appear to be a superficial challenge, relevant only to theory, Duchesne (2025) highlights the importance of sex/gender in empirical research, and lists the ways in which operationalisations of sex and gender in science impact people's everyday lives. The author calls attention to past scientific biases, which had given rise to claims about women's unsurpassable biological inferiority; notions of "innate differences" between male and female brains continue to allow for discrimination within scientific discourse, which is then usually targeted towards the group with a lower social standing (Fine, 2013, in Duchesne, 2025). Duchesne (2025) explains this as follows:

In human neuroscience, a majority of research into sex-related effects is derived from systematic comparison between groups of cis men and cis women (Rechlin et al. 2022). Neurobiological differences observed between male and female organisms are often assumed (unfalsifiably) to be the product of sexually driven evolutionary pressures caused by innate differences in sex-associated traits (Cahill 2006; Spets and Slotnick 2022). Under this biologically essentialist assumption, sex differences tend to be interpreted as the direct manifestation of sex-related biological mechanisms. Biologically essentialist assumptions of sex differences in the nervous system have also informed conceptualizations of gender identity, gender roles, and sexual orientation as directly resulting from the effect of gonadal hormones on brain development (Hines 2011; Roselli 2018). Decades of critical feminist research has challenged not only essentialist assumptions about biologically innate sex differences in the nervous system, but also the validity of a binary conceptualization as adequate operationalization of sex-related effects. Most importantly, critical neurofeminist analysis has been instrumental in exposing the harmful consequences for women, gender-diverse, and sexually marginalized populations of proposing unfalsifiable biologically essentialist neuroscientific explanations rooted in binary operationalization of sex (FaustoSterling 2019; Jordan-Young 2012; Llaveria Caselles 2021). (pp. 117)

Scientific research, especially that of neuroscience, has a big influence on popular beliefs in society, and presenting certain sex/gender differences as purely biological, while ignoring significant psychosocial factors, can further discriminate against minorities in society, especially in the political sense. In other words, those who control scientific discourse, very often impact people's everyday lives. The author also draws attention to the fact that resting sex and gender as two distinct and binary variables excludes several minorities, like intersex and transgender individuals, among others. For these reasons, the neurofeminist perspective on sex/gender is important, even outside of theoretical fields.

Duchesne (2025) further highlights how, apart from being misrepresented, women are also often forgotten in clinical research. She that "[d]espite the many contributions of neuroscience to the study of sex and gender, androcentric bias within the field has been

widely criticized in recent years for invisibilizing female biological realities and, therefore, compromising our ability to generalize experimental research findings to a broader population, including, in particular, women (de Lange et al., 2021; Shanksy and Woolley, 2016)” (pp. 116). Duchesne’s statements on neuroscientific research converge with analyses on the state of other scientific fields, which report that “the majority of research disciplines prefers male subjects” (Plevkova et al., 2020). In preclinical research, the exclusion of female participants is often justified by scientists and ethical committee board members, as they claim it could negatively affect women’s reproductive potential (Waltz et al., 2024). Other claims include the fact that female subjects pose more difficulties for researchers, as their oestrous cycles would have to be tracked to manage variability—claims, which have since been proven as untrue (Plevkova et al., 2020). It is worth mentioning that in the case of certain medical conditions, e.g. multiple sclerosis, it is the male sample that is misrepresented (Eccles, 2019). Scientific research, especially medical research, should aim for equality when representing samples, as basing the needs of the whole population on ungeneralizable findings can lead to misdiagnoses and ineffective treatments.

The sex/gender issue is also clear in the contexts of this thesis. Very often, researchers treat participant “sex” as a binary variable, whose nature is so obvious that its operationalisation does not even require explaining within the text. This phenomenon is clearly visible in the context of: the data about the interviewees in the corpus which this study depends on (see Chapter 2, Section 1); the various studies on how men and women use language (see Section 5.3); the research on links between depression and sex (see Section 6.1). As this particular study is a corpus analysis of linguistic markers identified within pre-existing scientific literature, the foundations of the thesis do not allow the author to approach the issue from the neurofeminist perspective; nonetheless, the issues plaguing research on sex/gender differences were kept in mind during research procedures, and are taken into consideration as possible limitations of the findings (see Chapter 3, Section 3).

6.1. Depression and the Sexes

The differences in depression between men and women run along two axes: the “how?”, and the “why?”. Regarding the former, there is a clear correlation between sex/gender and the epidemiology, phenomenology, and pathophysiology of depression (Altemus et al., 2014). Studies report that women are twice as likely to suffer from depression or other anxiety disorders throughout their lifetime (Kessler et al., 1994); the information about female

predisposition to depression is also included in the two major diagnostic manuals used worldwide (see Sections 4.2.1.1 and 4.2.2.1). Considering the latter, theorists propose three distinct hypotheses to explain women's preponderance in clinical depression: the impact of sex roles; the impact of social factors; and the impact of biological factors (Parker & Brotchie, 2010). What this means in practice, as a whole, is that the disease's onset, presentation, and cause vary between men and women.

A plethora of studies explore how depression symptoms manifest themselves in male and female patients. A study of 1500 outpatients with nonpsychotic MDD reported that women had a higher risk of comorbidities such as anxiety disorders, somatoform disorder, and bulimia, while men were more prone to alcoholism and drug abuse (Marcus et al., 2005). Other research shows that women are also more likely to report greater symptom severity, as well as to experience "increased appetite, weight, hypersomnia, interpersonal sensitivity and gastrointestinal symptoms" (Altemus et al., 2014, pp. 32). Additionally, women may experience fluctuation in their depression symptoms, which is linked to the different phases of their menstrual cycles (Haley et al., 2013 in Altemus et al., 2014). In turn, men display higher likelihood for suicidal ideation and higher suicide rates (Shi et al., 2021). When analysing coping strategy repertoires, Parker and Brotchie find that "in response to depression, the general finding is that women are more likely to internalize (e.g. go quiet, go to their bedroom, cry) and men are more likely to externalize (e.g. show anger, increase their alcohol intake)" (2010, pp. 434). This finding concurs with the Silencing the Self theory (see Section 6.2). Still, some researchers remain uncertain, and claim that gendered differences in depression may result from artefactual factors, such as the idea that women may be more likely to seek help than men, and provide more details during patient interviews. Such claims, however, have been discredited; Parker and Brotchie (2010) note that if the differences were of an artefactual nature, they would be significant at all stages of the disease's course, while research shows that the contrasts between genders are only predominant during the first episode of depression, and not recurrent ones. Furthermore, Parker and Brotchie (2010) highlight that these differences also emerge during public surveys, where help-seeking is not a relevant factor.

The question of why these differences between the sexes occur remains open. Researchers have considered social, behavioural, and biological factors. The first category, social factors, also encompasses environmental and cultural aspects. The rationale behind social theories is that gender roles in society, as well as gendered experiences, may affect an individual's perception of "the self", which then translates to disorder diathesis

(predisposition) (Parker and Brotchie, 2010). Some adverse experiences, such as childhood sexual abuse, occur more commonly to women (14.5%) than to men (7.2%) (Gorey and Leslie, 1997). This has led some researchers to extend the possible link between adverse childhood experiences and depression in adult-life, to an additional variable: gender (Altemus et al., 2014). Adding to this argument, in terms of behavioural factors research points to the possible influence of taught coping styles; studies have shown that during mid-childhood, boys express a tendency to develop avoidant coping styles, while girls demonstrate “preoccupied and anxious coping” (Carter, 2011 in Altemus et al., 2014, pp. 322). However, the strongest arguments appear to be linked to biological factors; researchers believe that social and cultural elements may simply function as modifiers of biology (Parker and Brotchie, 2010), as they can “alter gene expression and contribute to development of sex differences in brain and physiology throughout the lifespan” (Altemus et al., 2014, pp. 320).

The believed biological drivers of differences in depression between the sexes appear as early as the prenatal stage of life, and continue throughout puberty. In terms of the prenatal hypotheses, some claim that variation in exposure to androgens in utero may influence MDD susceptibility in adult life. Altemus et al. (2014) note that:

The prenatal sex difference in androgen exposures appears to have behavioral significance. Lower amniotic fluid testosterone levels at mid gestation (weeks 13–20) was associated with a negative response bias and less response to rewarding stimuli during a functional magnetic resonance imaging (fMRI) task among boys studied between 8 and 11 years of age (Lombardo et al., 2012). A negative response bias to affective tasks is a common finding in individuals with major depression (Jaworska et al., 2014). (pp. 321)

While these findings concerned only boys who were exposed to varied concentrations of androgens, it may be possible that the effects of such androgen deficits extend to girls as well, as they are less exposed to male gonadal hormones overall. In light of these findings, some speculate that certain qualities of androgens protect against depression, or at least ameliorate a portion of the disease’s symptoms. Furthermore, studies on maternal stress levels during pregnancy and lactation have also been linked to “more fearful and reactive behavior in female infants and children compared to male offspring” (Buss et al., 2012 and Grey et al., 2012 in Altemus et al., 2014, pp. 321). It may be that male offspring are better protected while female offspring are simultaneously more susceptible, substantially increasing the disparity noticeable in adult life experiences. Such findings indicate that the biological gestational predeterminants of adult affective disorders appear to disfavour women. However, I believe it is important to mention that the diathesis is not extendable to non-affective

neuropsychiatric disorders, such as autism, where the opposite findings have been reported; here, scientists believe that it is female offspring that are better biologically “protected” (Werling, 2016).

Strong arguments stand for the postnatal biological causes of sex differences in depression. Hypotheses often focus on the effects of puberty, and especially pubertal hormones, as it is during this period (approx. 12 years of age) that such major differences emerge (Hyde and Mezulis, 2020). During puberty, and all through the adulthood which follows, men experience little fluctuation in hormonal activity, while women are subject to intense monthly cycles of rises and drops in estrogen, progesterone, LH, FSH, oxytocin, prolactin, and more. It has been theorised that such hormonal instability is linked to a higher chance of mood dysregulation, and, therefore, possible dysfunction (Noble, 2005). Altemus et al. (2014) further speculate that the societal changes in women’s reproductive, such as bearing less children, may increase the detrimental effects of hormonal fluctuation; this is as “both pregnancy and lactation suppress hypothalamic–pituitary adrenal axis and autonomic responses to stress (Matthews and Rodin, 1992; Altemus et al., 1995; Entringer et al., 2010). This suppression state may have been the set point for anxiety and stress responsivity in women” (pp. 323). As evolution moves slowly, the fact that women are exposed to more cycles of menstruation than in previous centuries may be the reason behind their increased vulnerability to mood disorders.

6.2. Silencing The Self—The STS Theory

Despite the strength of biological arguments, approaching the issue from an intersectional perspective is of key importance. Quantitative research on hormones, genes, and brain reactivity may provide a strong foundation for research on sex differences in depression; however, it is important to remember that mental disorders, and especially mood disorders, have been heavily disputed and disagreed upon since the dawn of psychiatry (see Section 3). This perspective has been the result of their unquantifiable and highly idiosyncratic nature. For these reasons, a more personal approach to the issue of depression susceptibility in women has gained traction, giving rise to the “Silencing the Self” model (STS). As the model’s creator, Dana Jack, explains, “the model integrates aspects of attachment theory, relational theories, and cognitive theories of depression to explain women’s vulnerability to depression” (2010, pp. 7). Jack conducted a longitudinal study during which she asked women to describe their depression and the experiences which had led up to the disorder, thus redirecting the attention from society and its expectations to the voices of women.

In her breakthrough work, *Silencing the Self: Women and Depression* (1991), Jack ruminated on how in many cultures, women are pressured into repressing their negative emotions, especially anger, to satisfy expectations of “feminine goodness” and subordination (pp. 94). Jack further argued that these external expectations of women being “pleasing” and “unselfish” lead women to engage in self-silencing: the process of suppressing thoughts, emotions, and actions, which they believe could be met with disapproval from their partner. This self-silencing then gives rise to feelings of low self-worth, lack of agency, and entrapment. Jack identified four schemas of self-silencing: externalized self-perception, during which women judge themselves through external standards rather than their own values; care as self-sacrifice, which entails disregarding or going against one’s own needs with the belief that this is what is required in a successful relationship; silencing the self, the previously mentioned mechanism of censoring oneself to avoid conflict; and the divided self, which relates to the feeling of dissonance between inward experience and outside presence (2010, pp. 6). Based on these dimensions, Jack proposed a Silencing the Self Scale (STSS), which is a gender-neutral questionnaire used for assessing an individual’s level of self-suppression. The questionnaire and the theory’s framework have since been used in relation to conditions such as depression, various eating disorders, and premenstrual dysphoric disorder (Emran et al., 2020). The theory is linked to both causes, as well as manifestations of depression, which is why it may serve as a useful insight into possible linguistic differences between depressed men and women; if women are more likely to suppress emotions, it is possible that their language will be less emotive in terms of sadness, or filled with overcompensations and attempts at downplaying symptoms. While these specific examples are speculative, the cited literature strongly suggests the possible existence of linguistic differences between men and women dealing with depression.

Chapter Two

Methodology

Introduction

The aim of this chapter is to present the methodological background of the study, by describing the data sources, analytical tools, and procedures employed in the investigation of linguistic markers of depression. It is structured as follows. Section 1 of the chapter comprises information on the base corpus on which the study was conducted: the Distress Analysis Interview Corpus (Wizard of Oz) (DAIC-WOZ). The sub-sections detail the contents of the corpus, profile the interview participants, explain the interview protocol, and provide information on the psychological test, the eight item Patient Health Questionnaire (PHQ-8), which the participants completed prior to the interviews. Section 2 delves into the research questions and hypotheses of the thesis. The chapter finishes with Section 3, in which the procedures of compiling the final database and analysing the resulting corpus are presented. It contains sub-sections detailing the functioning of the Linguistic Inquiry and Word Count software (LIWC-22), as well as descriptions and justifications of the linguistic markers of depression chosen for analysis in this thesis. The chapter ends with a step-by-step description of the dataframe construction process.

1. The Source Corpus

The present study was conducted on materials from the Distress Analysis Interview Corpus Wizard of Oz (DAIC-WOZ), which contains 189 recorded and annotated semi-structured interviews (Gratch et al., 2014; DeVault et al., 2014). DAIC-WOZ is a multimodal sub-corpus of the larger DAIC dataset. Created with the intention to better understand the development of anxiety and depressive disorders, the corpus contains interview audio, transcripts, physiological data, and depth sensor positional data, which, when analysed, aim to ease the diagnosis of such mental conditions. The interviews were collected as part of a wider effort to develop a computer agent capable of helping the medical field in the diagnosis process. Access to the database was acquired from the USC Institute for Creative Technologies, upon prior application (dcapswoz.ict.usc.edu/, DOA: 30.11.2025).

While DAIC contains four different interview types, half of which are human-human interactions (face-to-face and teleconference), and half human-computer (automated and

Wizard of Oz), the DAIC-WOZ contains only the Wizard of Oz variant (WoZ). The WoZ condition of interviews is led by a computer agent avatar called Ellie, who is controlled by a human sitting outside of the room. This condition of the experiment is a middle ground, where interviewees see only the virtual agent, and are presented two varying framing conditions, wherein one reveals the human interviewer, and the other does not. The Ellie agent is part of the SimSensei Kiosk project developed by DeVault et al. (2014), and was designed with the purpose of creating a comfortable conversational scenario, during which participants can safely share personal experiences. The agent's second purpose is to "create interactional situations favorable to the automatic assessment of distress indicators, defined as verbal and non-verbal behaviours correlated with depression, anxiety or post-traumatic stress disorder (PTSD)" (DeVault et al., 2014, pp. 1061). Developing a virtual agent capable of aiding the mental health diagnosis process has become a major goal at the crossroads of medical and computer sciences (Zou and Topol, 2025). With the growing capabilities of Large Language Models and Artificial Intelligence, it has been thought that such tools can benefit professionals in the field; raw data provided by patients could possibly be used to diagnose or prognose certain diseases (Gorenshtein et al., 2025), or, as in the case of this thesis, be used for identifying linguistic markers.

1.1. Participants

For the whole corpus, samples from two populations of participants were drawn, one being veterans of the U.S. Armed Forces, the second being the general public. Members of both populations were residing in California at the time of the interviews taking place. Representatives of the general public were recruited online through the American listing website, Craigslist.com, while veterans were contacted at the facility of US Vets, a Californian non-profit organisation. Importantly, only the general public sample took part in the Wizard of Oz condition, bringing the total number of interviewed participants to $N = 193$. Each participant completed a set of psychiatric tests aimed to detect signs of anxiety, depression, and PTSD (see Section 1.2. below).

1.2. Interview Protocol

In the WOZ condition, participants would be placed alone in a room and sat in front of a computer screen, on which Ellie was launched. Two researchers were responsible for guiding Ellie's behaviour in conversation; one controlled non-verbal behaviour, while another controlled vocal utterances. The WOZ interviews lasted from 5 to 20 minutes each. Conversation protocol guidelines were put in place, however, minor changes were made per participant, due to the nature of interaction; this was necessary, as the conversation aimed to be as natural as possible. The agent began by asking neutral control questions, and gradually progressed to asking more symptom- and experience-targeted questions. The interviews were later transcribed using the ELAN software (Brugman and Russel, 2004). Gratch et al. (2014) present the following as an example WOZ interview fragment:

“Ellie: Who's someone that's been a positive influence in your life?

Participant: Uh my father.

Ellie: Can you tell me about that?

Participant: Yeah, he is a uh

Participant: He's a very he's a man of few words

Participant: And uh he's very calm

Participant: Slow to anger

Participant: And um very warm very loving man

Participant: Responsible

Participant: And uh he's a gentleman has a great sense of style and he's a great cook.

Ellie: Uh huh

Ellie: What are you most proud of in your life?” (pp. 3124)

As can be seen above, boundaries between utterances were put in place when participants hesitated and fell silent for 300 milliseconds or more. For this reason, transcription data also contained information about utterance duration (utterance start time and utterance finish time). As the data was collected by means of conversation, participants often replied in fragmented statements, instead of fully grammatical sentences; interjections, such as “uh”, were also included within the transcripts.

1.3. Psychological Testing: PHQ-8

As the interview data was collected with the purpose of better understanding depression, anxiety, and PTSD, Gratch et al. (2014) also collected data on psychological wellbeing, using

the Patient Health Questionnaire-8 (PHQ-8) (Kroenke et al., 2002). It has been reported that the PHQ-8 is “more convenient to use than a diagnostic algorithm” (Kroenke et al., 2009), especially when applied in large clinical studies, such as the DAIC-WOZ corpus. Importantly, the PHQ-8 questionnaire is based on DSM-IV diagnosis criteria for depression, and not on the latest updates from the DSM-V manual. The PHQ-8 asks the following question: “Over the last two weeks, how often have you been bothered by any of the following problems?”, and consists of eight items:

1. Little interest or pleasure in doing things;
2. Feeling down, depressed, or hopeless;
3. Trouble falling or staying asleep, or sleeping too much;
4. Feeling tired or having little energy;
5. Poor appetite or overeating;
6. Feeling bad about yourself - or that you are a failure or have let yourself or your family down;
7. Trouble concentrating on things, such as reading the newspaper or watching television;
8. Moving or speaking so slowly that other people could have noticed? Or the opposite - being so fidgety or restless that you have been moving around a lot more than usual. (Kroenke et al., 2009, pp. 171)

For each item, a participant must respond with a value between 0–3, where: 0 corresponds to “Not at all”; 1 corresponds to “Several days”; 2 corresponds to “More than half the days”; 3 corresponds to “Nearly every day”. The participant scores from the eight items are summed, with the maximum score of 24 points. Kroenke et al. (2009, pp. 164) report that “a PHQ-8 score of ≥ 10 , has an 88% sensitivity and 88% specificity for major depression”. According to these test guidelines, the DAIC-WOZ corpus provides a PHQ-8 binary score per participant, with participants scoring less than 10 points being classified as “not depressed”, and 10 or more as “depressed”. For a fraction of participants, scores for each test item are also made available.

2. Research Questions and Hypotheses

Informed by the literature described and analysed in Chapter One, the current work hopes to answer two complementary research questions concerning the connections between language, depression, and gender.

RQ1: *Do symptoms of depression manifest themselves in patient language in the form of linguistic markers?*

This question primarily serves as a benchmark for the development of RQ2. It has been explored in prior literature, and has been gaining particular traction since the boom of Artificial Intelligence and Large Language Models in scientific research (Omar and Levkovich, 2025). Theoretical approaches suggest that depressed and non-depressed people may exhibit linguistic differences due to the fact that depression impacts an individual's mental wellbeing and cognitive functioning. Therefore, a depressed individual may exhibit behaviours such as lowered mood, anxiety and negative affect, as well as experience impeded attention, impaired memory, and slower processing speed (Perini et al., 2019). In addition, empirical studies have found concrete links between certain linguistic markers and depression (Eichstaedt et al., 2018; Newell et al., 2018; Zimmerman et al., 2017; Tackman et al., 2019; Smirnova et al., 2018). For an extensive overview of available literature, refer to Savekar et al. (2022). Posing this research question establishes a benchmark against which gender-dependent effects can be compared.

RQ2: *Do differences in the epidemiology, phenomenology, and pathophysiology of depression between men and women translate to differences in linguistic marker presence?*

The second question constitutes the novelty value of the present study; to the extent of my knowledge, no research has been conducted on the influence of gender on the linguistic markers of depression. However, literature on linguistic profiles proposes the existence of the genderlect, a style of speech unique to members of a particular gender (Tannen, 1990). As a baseline, this suggests that men and women may be characterised by their linguistic choices, even in contexts unrelated to depression. To add to this, a strong body of literature on depression highlights the differences in disease course and symptoms between men and women; as presented in Chapter One (see Section 6), women are not only more likely to suffer from depression, but are also more susceptible to a wider array of depression symptoms. These include aspects such as weight or appetite fluctuation, and heightened interpersonal sensitivity, which are additionally dependent on the phases of the menstrual cycle (Altemus, 2014). Socio-psychological theories, like the Silencing the Self theory (see Chapter One, Section 6.2.), additionally postulate that women are also exposed to different social experiences, which alter their conscious and subconscious behaviour. While the majority of studies on sex-dependent language differences are observational, some empirical, quantitative literature exists (Mulac et al., 1988; Newman et al., 2008; Xia, 2013; Merrill et

al., 2015). Based on the available theoretical and empirical literature, I test whether these findings extend to sex-dependent linguistic differences in depression.

With regard to RQ1, I hypothesise that:

H1: Depressed participants will exhibit different behaviours in terms of language use in comparison to healthy participants.

This broad hypothesis branches out into three precise hypotheses, which target specific linguistic markers of depression:

H1a: Depressed participants will use more negatively valenced language when compared to non-depressed participants.

H1b: Depressed participants will use more First Person Singular Pronouns when compared to non-depressed participants.

H1c: Depressed participants will use more Absolutist words when compared to non-depressed participants.

For concrete operationalisations of the chosen linguistic markers of depression, see Section 3.1.1.

With regard to RQ2, I hypothesise that:

H2: Depressed men and women will differ between themselves in terms of used linguistic markers of depression.

While H2 concerns the same markers as H1 (negative valence, FPSP, absolutist words), it is non-directional, due to lack of previous literature on the effect of gender on linguistic markers of depression.

3. Present Study: Methods

The present study adopts a quantitative corpus-based approach to the analysis of linguistic markers of depression. Participant transcripts were extracted from the DAIC-WOZ corpus, pre-processed and compiled into a master database, and later analysed using the Linguistic Inquiry and Word Count software (LIWC-22).

3.1. Linguistic Inquiry Word Count

The Linguistic Inquiry and Word Count (LIWC) is a software developed for the general purpose of analysing language, created by Pennebaker et al. (1999). As stated by one of the creators, “[o]ur goal was to create a program that simply looked for and counted words in psychology-relevant categories across multiple text files” (Tausczik & Pennebaker, 2010, pp. 27). To achieve this, LIWC was programmed to include a range of dictionaries, each containing words meant to reflect various cognitive states; for example, a dictionary may list words linked to “negative emotion”. The programme analyses input texts word by word, and checks whether each word is present within the user-selected dictionaries. The final output is a percentage, reflecting how many words within a text belong to a given dictionary category. Categories are structured hierarchically, as texts can be analysed on different levels; if a text contains the word “you”, the consequent categories of “you”, “second person personal pronoun”, “pronoun”, and “function word” would be marked as present. Moreover, words are also not restricted to being present in only one dictionary, and can match more than one category at a time. Significantly, the LIWC user manual highlights that analysis quality improves as word count increases, suggesting that a lower boundary of 25–50 words should be set as a cut off point for reliability (www.liwc.app/help/howitworks, DOA 25.04.2026).

LIWC (specifically LIWC-22) was chosen as the analytic tool in the present study, due to it being an established software for conducting research on the linguistic markers of psychological states; in a systematic literature review on linguistic markers of depression, Savekar et al. (2022) reported on 25 studies, 18 of which had used LIWC as the primary analytic method (see Chapter One, Section 5). As stated by Koehler et al. (2018), LIWC is an appropriate tool for such tasks, as the programme “offers explanatory power and transparency, and lends itself to hypothesis driven models for detecting depression” (pp. 3). LIWC has also been previously used for analyses conducted on the DAIC and DAIC-WOZ corpora, either as the primary method, or as a baseline (Zhang & Xiao, 2020; Villatoro-Tello et al., 2021; Chopra et al., 2025).

3.1.1. Markers and Dictionaries

A range of linguistic markers of depression have been identified in previous literature, both in terms of verbal and non-verbal behaviour (see Chapter One, Section 5). For the purpose of this analysis, three main categories of text-only marker categories were specified: emotion valence, pronoun use, and absolutist word use. The selection of these categories is grounded

in prior linguistic research on depression. The emotional valence of language is strongly linked to psychological research on depression, such as Beck's cognitive model of depression (1967). The model, constructed on the basis of 966 evaluated psychiatric patients, noted that there exists a strong link between depression and negative affect, linked to both the past and the future, especially when targeted towards the self (Beck, 2002, pp. 34). With this in mind, it may be expected that depressed patients will use language linked to negative affect more often than the healthy controls (Newell et al., 2018). The second category of pronouns, further builds upon the cognitive model of depression; empirical research has shown that depressed individuals use significantly more First Person Singular Pronouns (FPSP) in comparison to healthy controls (Newell et al., 2018; Zimmerman et al., 2017; Huang et al., 2024). This is aligned with the idea that depressed patients exhibit a tendency to "look inwards", focusing on the self instead of on the outside world. The final marker category targeted in this study is the absolutist word category, based on findings by Al-Mosaiwi and Johnstone (2018a). Absolutist words, such as "every", "never", or "definitely", reflect a bias to think in extreme, binary categories; in an extensive study of online forums, the researchers found that forums related to depression, anxiety, and suicide, were more likely to contain absolutist words than control forums. Taken together, as the three chosen marker categories have been found to possess the potential to serve as differentiating linguistic markers of depression, they may reliably be later used to look for gender differences in depression.

To test for the chosen markers, two LIWC dictionaries were activated during analysis; the internal LIWC-22 English dictionary and the Absolutist Dictionary created by Al-Mosaiwi and Johnstone (2018b) (DOA: 03.04.2025). Dictionaries are the basis of LIWC analyses, as they match words present in a text to linguistic or psychometric tags; to ensure the reliability and accuracy of the LIWC-22 English dictionary, Boyd et al. (2022) followed a seven step process of dictionary construction. The phases consisted of 1) word collection, where the researchers drew examples from previous LIWC dictionaries, psychological metric scales, and thesauri; 2) judge rating, during which a team of human raters determined whether each word was appropriate for a category; 3) base rating, where the frequency of each word's appearance in a large corpus was calculated to check word significance; 4) candidate word list generation, during which insignificant constituents were omitted; 5) psychometric evaluation, during which internal consistency statistics were calculated for each category; 6) refinement, which consisted of re-running steps 1–5 once again; 7) the addition of four summary variables, previously researched by the creators of LIWC (Boyd et al., 2022, pp. 6–7). The resulting LIWC-22 English dictionary is extensive and contains various marker categories.

For the purpose of the present study, this wide selection was narrowed down to the relevant categories. The chosen LIWC tags are (for constituent word examples for each marker, see Table 1. below):

- To test for the valence of emotional language, I used the in-built LIWC markers: Positive Emotion, Negative Emotion, Anxiety, Anger, Sadness, Positive Tone, and Negative Tone.
- To assess the frequency of pronoun use, I used the in-built LIWC markers: Total Pronouns, Personal Pronouns, and Impersonal Pronouns. “Personal Pronouns” is an umbrella category, which contains sub-categories for each specific pronoun: First Person Singular, First Person Plural, Second Person, Third Person Singular, and Third Person Plural.
- To test for absolutist language, I used the “Absolutist” marker category from the LIWC Absolutist dictionary (Al-Mosaiwi and Johnstone, 2018b).

The LIWC Absolutist dictionary is narrower in comparison to the LIWC-22 English dictionary. Initially, the researchers differentiated between two word classes: absolutist and nonabsolutist words, which “indicate magnitudes or probabilities; absolute words do so without nuance (i.e., always, totally, entire), whereas nonabsolute words indicate some degree of nuance (i.e., rather, somewhat, likely)” (Al-Mosawi and Johnstone, 2018a, pp. 532). The dictionary applied in the present study contained only the former category: Absolutist words. The process of this dictionary’s construction was also refined to ensure reliability, consisting of multiple phases: 1) brainstorming, during which the researchers created a list of 500 absolutist and nonabsolutist words; 2) piloting and cleaning, where uncommon and obscure words were removed; 3) judge rating, during which five independent judges, two of whom were clinical psychologists, took majority votes to decide which words should remain within each category; 4) final cleaning, where, based upon the previous voting, a master list of 19 absolutist words was refined (Al-Mosawi and Johnstone, 2018a, pp. 532).

Table 1. A table containing linguistic marker categories and their example constituent words from the LIWC-22 English dictionary and the Absolutist word dictionary. Constituents followed by an asterisk indicate that all word suffixes were accounted for, i.e. frustr*: frustrated, frustrating, frustration, and more. LIWC-22 English

dictionary examples were sourced from Pennebaker et al. (1999) and Boyd et al. (2022). The list of example absolutist words was retrieved from Al-Mosaiwi and Johnstone (2018a).

LIWC Marker	Marker Tag	Example constituents
Positive Emotion	<i>emo_pos</i>	Happy, pretty, good, love, hope
Negative Emotion	<i>emo_neg</i>	Hate, worthless, enemy, bad, tired
Anxiety	<i>emo_anx</i>	Nervous, afraid, tense, worry, fear
Anger	<i>emo_ang</i>	Hate, kill, pissed, mad, angry, frustr*
Sadness	<i>emo_sad</i>	Grief, cry, sad, disappoint*
Positive Tone	<i>tone_pos</i>	Good, well, new, love
Negative Tone	<i>tone_neg</i>	Bad, wrong, too much, hate
Personal Pronouns	<i>ppron</i>	I, you, my, me
First Person Singular	<i>I</i>	I, me, my, myself
First Person Plural	<i>we</i>	We, our, us, lets
Second Person	<i>you</i>	You, your, yourself
Third Person Singular	<i>shehe</i>	He, she, her, his
Third Person Plural	<i>they</i>	They, their, them, themselves, themselves
Impersonal Pronouns	<i>ipron</i>	That, it, this, what
Absolutist words	<i>Absolutist</i>	Absolutely, all, always, complete*, constant*, definitely, entire, ever, every, everyone, everything, full, must, never, nothing, totally, whole

3.2. Final Database Compilation and Data Processing

To compile the final database from the source corpus, the pandas Python library (McKinney, 2010) was used for data cleaning and organisation. The database compilation was executed in two stages: 1. Extracting the relevant data; 2. Application of LIWC to the extracted data. The initial stage consisted of extracting the necessary data from the DAIC-WOZ corpus. Each participant folder was downloaded from the corpus directory, however, only the participant interview transcript files were extracted. Each transcript file contained four columns: start

time, end time, speaker, and value. As utterance length was not a dependent variable in this study, the two former columns were removed. Next, only the values (utterances) linked to the “Participant” speaker were retained; as the analysis was quantitative and not qualitative, the Ellie agent’s utterances were not necessary for any context analysis. Furthermore, keeping only the Participant utterances guaranteed that the analysed language was produced either by depressed or not depressed individuals, and not by the experimenters guiding the Ellie agent. The DAIC WOZ corpus provides 189 participant interview files, however, not every participant file contains data on participant depression status and participant gender; as these values are of key importance to the present study, only the fully-described participants were included (N = 142, Nmale = 79, Nfemale = 63). Among the remaining participants, 42 constituted depressed individuals, and 100 non-depressed individuals (Ndep/male = 18, Nnon-dep/male = 61, Ndep/female = 24, Nnon-dep/female = 39). When only relevant participants remained, all transcripts were joined into one master file, containing four columns: participant, whole transcript, gender, and PHQ-8 binary value.

The second stage of database compilation consisted in running the cleaned master file through the chosen LIWC dictionaries (see Section 3.1.1. above). The LIWC software appended the calculated quantity percentages per marker, per participant, resulting in the final master database. Due to the DAIC WOZ copyright, the original master database cannot be made available. Access to the full DAIC WOZ can be applied for under the following web address: dcapswoz.ict.usc.edu/ (DOA: 27.04.2026). However, an anonymised version of the master database, has been made available here: github.com/antonijordan/GenderAndDepression (DOA: 01.06.2026). The anonymised version contains the Participant ID, LIWC marker output scores, PHQ-8 Binary score, and a binary Gender value.

Chapter Three

The Analysis

Introduction

Chapter Three presents the results of the conducted analyses and discusses the implications of the findings for research on automated depression detection, the use of linguistic markers in clinical contexts, as well as gender-dependent linguistic differences. The results are discussed in light of the empirical and theoretical backgrounds of Affective Computing, Depression-Oriented Corpus Linguistics and Gender Studies. Further, the chapter acknowledges possible limitations of the present study, and concludes with proposals of future research directions.

1. Results

All statistical analyses were coded and executed in Python; the code can be accessed under the following address: github.com/antonijordan/GenderAndDepression (DOA: 01.06.2026).

To test whether depressed and non-depressed people use language differently (Hypotheses 1: 1a, 1b, 1c), a multi-method analytic approach was employed, consisting of inferential group comparisons and a logistic regression model. As an introductory analysis, independent samples t-tests and Mann-Whitney U tests were run, to see whether there were any significant differences between depressed and non-depressed samples when each marker (from the negative valence, pronoun use, and absolutist words categories) was taken into consideration separately (see Table 2.). Data normality for each marker was first tested using the Shapiro-Wilk test, to ensure that the correct parametric or non-parametric test was used. Due to the high number of comparisons (16 features), p -values were corrected by the Benjamini-Hochberg False Discovery Rate (FDR) to avoid Type I errors. This procedure was chosen instead of Bonferroni correction, as it is less conservative, and allows for more exploratory approaches in studies based on corpus data (Lee and Lee, 2018). Both raw p -values, and FDR-corrected p -values are provided. Effect size was calculated as Cohen's d , where $d=0.2$ is understood as small effect size, $d=0.5$ as medium, and $d=0.8$ as large.

Table 2. A summary table presenting the independent samples t-tests or Mann-Whitney U tests comparing the prevalence of linguistic marker use between depressed and non-depressed samples.

RQ1 Results												
Group differences (Non-Depressed vs Depressed)												
Marker	column	Test	Statistic	p (raw)	Cohen's d	Mean (Non-Dep)	SD (Non-Dep)	Mean (Dep)	SD (Dep)	p (FDR)	sig_fdr	sig
Anxiety	emo_anx	Mann-Whitney U	1,283.000	0.0000	0.658	0.122	0.149	0.238	0.199	0.0030	True	**
Negative tone	tone_neg	Mann-Whitney U	1,354.500	0.0010	0.602	1.767	0.735	2.254	0.879	0.0070	True	**
Negative emotion	emo_neg	Mann-Whitney U	1,504.000	0.0080	0.518	1.161	0.567	1.522	0.808	0.0420	True	*
2nd person (you)	you	Mann-Whitney U	2,503.000	0.0720	-0.413	1.665	1.425	1.184	0.825	0.1650	False	
Positive tone	tone_pos	Mann-Whitney U	2,528.000	0.0560	-0.385	4.669	1.357	4.182	1.165	0.1650	False	
Sadness	emo_sad	Mann-Whitney U	1,659.500	0.0490	0.348	0.215	0.172	0.284	0.220	0.1650	False	
Anger	emo_anger	Mann-Whitney U	1,824.500	0.2190	0.330	0.299	0.230	0.410	0.416	0.3500	False	
Positive emotion	emo_pos	Mann-Whitney U	2,515.500	0.0640	-0.321	2.162	0.845	1.892	0.840	0.1650	False	
1st person singular (I)	i	t-test	-1.634	0.1040	0.296	9.816	1.828	10.378	1.963	0.2090	False	
Impersonal pronouns	ipron	t-test	-1.254	0.2120	0.231	6.086	1.298	6.385	1.284	0.3500	False	
Pronouns (total)	pronoun	t-test	-0.860	0.3910	0.162	19.600	2.944	20.051	2.626	0.4820	False	
3rd person plural (they)	they	Mann-Whitney U	1,887.000	0.3420	0.155	0.673	0.456	0.747	0.497	0.4560	False	
Absolutist words	Absolutist	Mann-Whitney U	1,880.500	0.3280	0.138	0.844	0.444	0.905	0.438	0.4560	False	
1st person plural (we)	we	Mann-Whitney U	2,263.500	0.4660	-0.108	0.498	0.412	0.454	0.419	0.5330	False	
Personal pronouns	ppron	t-test	-0.373	0.7100	0.070	13.512	2.287	13.665	2.111	0.7570	False	
3rd person singular (she/he)	shehe	Mann-Whitney U	2,102.500	0.9930	-0.020	0.767	0.629	0.755	0.571	0.9930	False	

Out of sixteen tested linguistic markers, three proved to be statistically significant after FDR correction: Anxiety ($M1 = 0.122$, $SD1 = 0.149$ vs. $M2 = 0.238$, $SD2 = 0.199$; $p_{\text{raw}} = <0.0001$, $p_{\text{FDR}} = 0.003$, $d = 0.658$), Negative tone ($M1 = 1.767$, $SD1 = 0.735$ vs. $M2 = 2.254$, $SD2 = 0.879$; $p_{\text{raw}} = 0.001$, $p_{\text{FDR}} = 0.007$, $d = 0.602$), and Negative emotion ($M1 = 1.161$, $SD1 = 0.567$ vs. $M2 = 1.522$, $SD2 = 0.808$; $p_{\text{raw}} = 0.008$, $p_{\text{FDR}} = 0.042$, $d = 0.518$). These three marker categories are all linked to lowered or depressed mood, and represent the strongest effects within the dataset. The effect size of each marker is considerable (medium), with depressed individuals using more of these markers than the non-depressed sample.

Within the negative emotion category, the Sadness marker appeared to be statistically significant after the first stage of testing ($M1 = 0.215$, $SD1 = 0.172$ vs. $M2 = 0.284$, $S2 = 0.220$; $p_{\text{raw}} = 0.049$, $d = 0.348$); however, the result did not stand after FDR correction ($p_{\text{FDR}} = 0.165$). This suggests that although depressed participants used more words linked to sadness, the result should be treated with caution. Absolutist words, associated with cognitive rigidity and extreme polarity, have also been strongly correlated with depression and suicidal ideation in previous studies, forming the basis of Hypothesis 1c; however, in this analysis, they were found to be insignificant ($p_{\text{raw}} = 0.328$, $p_{\text{FDR}} = 0.456$, $d = 0.138$).

The markers from the positive category, Positive Emotion ($p_{\text{raw}} = 0.064$) and Positive Tone ($p_{\text{raw}} = 0.056$), were on the verge of being considered significant after initial Mann-Whitney U tests, yet failed to withstand FDR correction (tone_pos: $p_{\text{FDR}} = 0.1650$; emo_pos: $p_{\text{FDR}} = 0.1650$). While, due to its lack of statistical significance, the result can only be seen as a possible tendency within the dataset, it appears to confirm assumptions made in previous literature. The non-depressed sample (tone_pos: $M1 = 4.669$, $SD1 = 1.357$; emo_pos: $M1 = 2.162$, $SD1 = 0.845$) used more positive language than the depressed sample (tone_pos: $M2 = 4.182$, $SD2 = 1.165$; emo_pos: $M2 = 1.892$, $SD2 = 0.840$), with a small-to-medium effect size (tone_pos: $d = -0.385$; emo_pos: $d = -0.321$).

Surprisingly, no effects related to pronoun use were found in this analysis, despite strong links between depression and the First Person Singular Pronoun (FPSP) forming the basis of Hypothesis 1b and being well-documented in previous literature. Although the effect for the “I” pronoun was in the expected direction ($d = 0.296$), consistent with theories of increased self-focused language in depression, it did not reach significance in this sample ($p_{\text{raw}} = 0.1040$, $p_{\text{FDR}} = 0.2090$). While also insignificant, the Second Person “You” Pronoun reported the lowest p-value (raw $p = 0.072$); however, this did not withstand FDR correction ($p_{\text{FDR}} = 0.1650$). The “you” pronoun appeared to be more present in the language of the non-depressed sample ($M1 = 1.665$, $SD1 = 1.425$, $d = -0.413$), than the depressed sample ($M2 = 4.182$, $SD2 = 1.165$), which is also in line with the theories regarding depressed language being more self-focused. This is discussed further in the discussion section (see Section 2). For a visual representation of the results described above, see Figure 1.

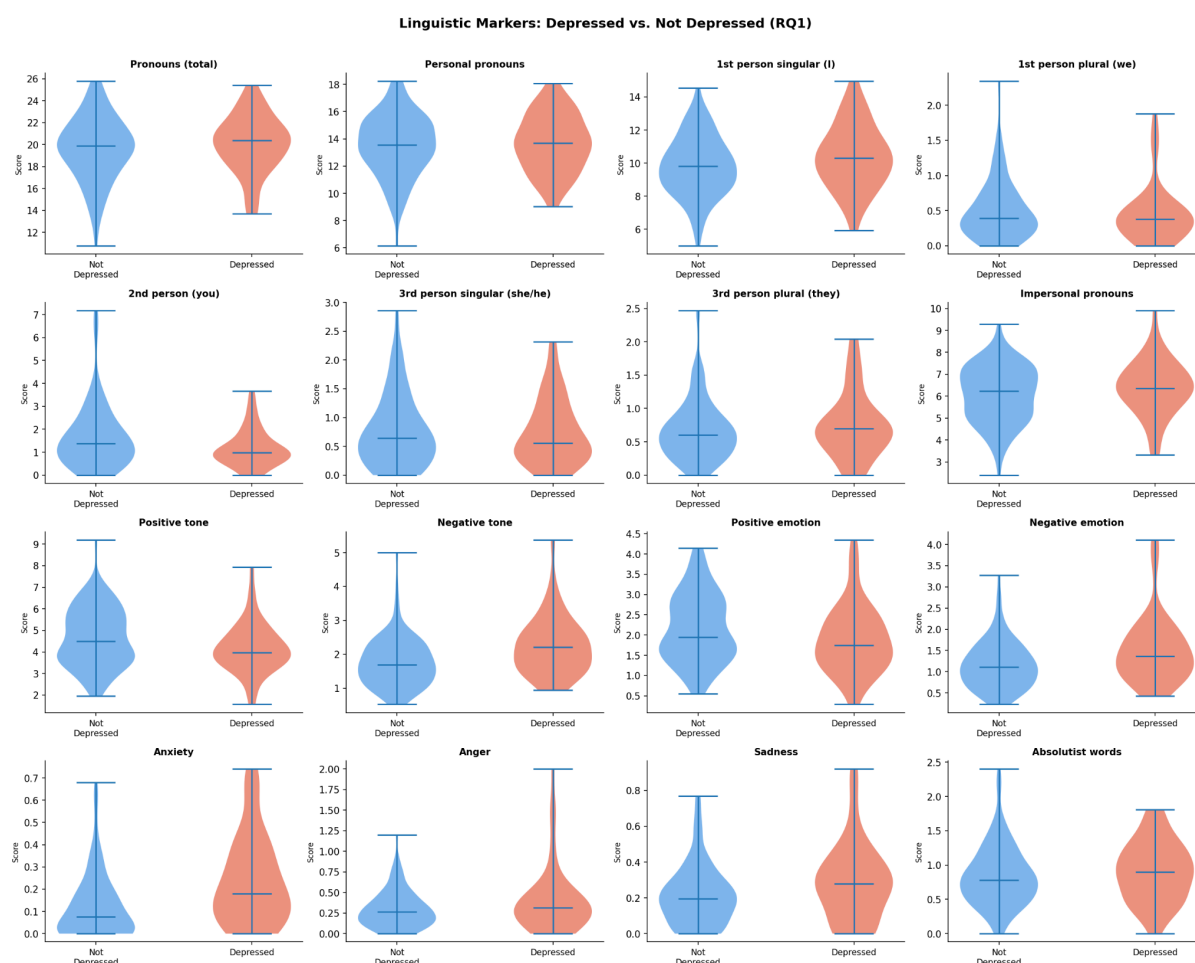


Figure 1. Violin plots presenting the distribution of differences between the non-depressed (blue) and depressed (red) samples per linguistic marker. The Anxiety, Negative Tone, and Negative Emotion markers were statistically significant.

In the second round of analyses for Hypothesis 1(1a, 1b, 1c), a logistic regression model was applied to the data, in order to see whether different results emerged when all markers were considered simultaneously. Two markers, Pronouns Total and Personal Pronouns, were removed from the analysis. This is as they are overarching categories which include many pronoun sub-markers, e.g. the FPSP. Furthermore, correlation matrices were calculated, revealing high correlation values (0.78 and 0.9 respectively) between two marker pairs: negative emotion with negative tone and positive emotion with positive tone. To reduce the risk of multicollinearity, negative emotion and positive emotion markers were also removed from further analyses. Tone markers were kept within the data, as within the LIWC software, they reflect a ratio of positive and negative language used by a person, in contrast to emotion markers, which provide a raw frequency of positive words and negative words; for this reason, tone is more informative within the present study (www.liwc.app/help/liwc#Tone,

DOA 27.05.2026). The remaining markers additionally underwent Variance Inflation Factor (VIF) testing, to ensure that no variable overlap occurred within the regression model; all of the twelve markers exhibited a VIF value of <3 , suggesting minimal inflation levels, and passing necessary safety benchmarks. The model demonstrated modest discriminative ability, with a cross-validated AUC value of 0.687 (SD = ± 0.089), meaning that it scored above chance, but below a threshold of strong classification confidence. Importantly, the model was strongly biased towards non-depressed samples in terms of recall (non-dep: 0.93, dep: 0.52); this is most likely caused by the imbalances of sample size, where depressed participants were under-represented (non-dep $n = 100$, dep $n = 42$). Regardless of the skewed sample sizes, the model reported a macro-averaged F1 score of 0.75, and weighted F1 score of 0.80. So, while not without flaw, these metrics can be telling when used for an exploratory analysis, such as the one detailed in this thesis (Hinojisa Lee et al., 2024).

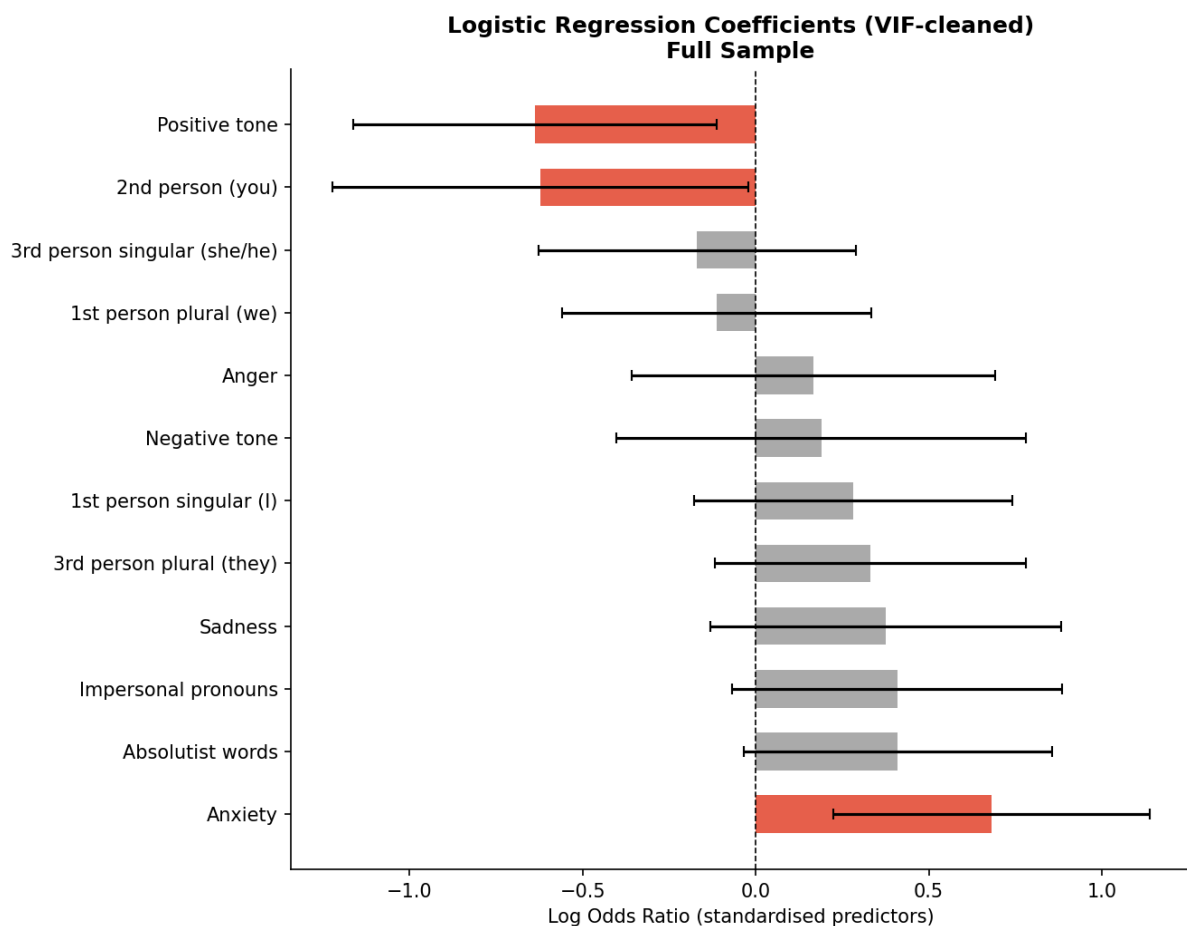


Figure 2. Logistic regression coefficients for linguistic markers ($n = 12$) when predicting from the full sample of participants ($n = 142$). Log odds ratios are presented with 95% confidence intervals. Positive values indicate a higher risk of depression, while negative values indicate lower risk. Statistically significant markers ($p < 0.05$) are highlighted in red. Predictors are standardised.

As can be seen in Figure 2., the logistic regression model found three linguistic markers of depression to be statistically significant: Positive Tone, the Second Person “You” Pronoun, and Anxiety. Positive Tone was a negative predictor of depression (Odds Ratio, OR = 0.53, 95% CI [0.31, 0.89], $p = .017$). Similarly, Second Person Pronoun use was also a negative predictor of depression (OR = 0.54, 95% CI [0.29, 0.98], $p = .042$), meaning that depressed people used less language focused on the “other” than on the “self”. This is complementary to results of increased FPSP use noted in other studies, which also confirm more self-oriented language; however, this was not confirmed statistically by this given model. Anxious language was a positive predictor of depression (OR = 1.98, 95% CI [1.25, 3.12], $p = .004$). This result indicates that higher use of anxious language was associated with roughly twice the odds of depression. Several additional markers showed trends in theoretically expected directions, including negative tone (OR = 1.20), sadness words (OR = 1.46), and absolutist language (OR = 1.51), though these did not reach significance, possibly given the model’s modest statistical power.

When combined, the two statistical analyses exhibit some convergence. The sole marker which emerged as significant in both inferential group comparisons and in logistic regression was Anxiety. This is a strong convergence, which provides robust evidence that anxious language can reliably be used to identify whether an individual is exhibiting depressive symptoms. The Negative Tone marker was a strong indicator of depression in the independent Mann-Whitney U analysis, and also showed a trend in this direction in logistic regression, despite not achieving statistical significance; this means that when controlling the effects of all markers simultaneously, words linked to negative emotions no longer successfully served as stand-alone predictors of depression. Complementarily, although the Positive Tone marker was not significant in independent analyses, it was a strong negative marker of depression. These two findings fit theoretical literature, which suggests that depression is linked to increased negative thought and affect (Beck, 1967). Finally, while the Second Person “You” Pronoun did not pass the FDR correction in the initial analyses, it later proved meaningful in logistic regression, as a negative predictor; it may be worth taking into consideration in future analyses. Taking both analyses into consideration, Hypothesis 1 was partially confirmed: to an extent, depressed and non-depressed participants use different language. More specifically, Hypothesis 1a was partially confirmed, with negative valence markers, particularly Anxiety and Negative Tone, proving significant. Hypotheses 1b and 1c were not confirmed, as neither First Person Singular Pronoun use nor Absolutist Words

reached significance after FDR correction. However, it is important to understand that these linguistic markers are not enough to successfully diagnose an individual, especially as the results of this thesis vary from the results of previous literature; this will be further discussed in Section 2.

A multi-method statistical approach was also applied to test whether depressed men and women differed in language use (Hypothesis 2). Similarly to the initial RQ1 analyses, independent samples t-tests and Mann-Whitney U tests were run to assess the influence of each linguistic marker when their effects are considered separately (see Table 3.). Initial p values also underwent FDR correction, due to small sample sizes; out of 16 linguistic markers, none survived FDR correction. Prior to FDR correction, the FPSP marker almost achieved statistical significance ($M1 = 10.683$, $SD1 = 1.544$ vs. $M2 = 9.730$, $SD2 = 2.301$; $p_{\text{raw}} = <0.063$, $p_{\text{FDR}} = 0.455$, $d = -0.578$). The negative Cohen's d score suggests that FPSP is more prevalent in depressed women than in depressed men; this result may be in line with Silencing the Self theories (see Chapter One, Section 6.2.). However, it is important to address that this result should be treated as a possible trend, due to its diminished significance upon correction. For a visual representation of the results described above, see Figure 3.

Table 3. A summary table presenting the independent samples t-tests or Man-Whitney U tests comparing the prevalence of linguistic marker use between depressed women and depressed men.

RQ2 Results									
Independent sample tests: Depressed Women vs. Depressed Men									
Marker	Test	Mean (Female)	SD (Female)	Mean (Male)	SD (Male)	Statistic	p (raw)	p (FDR)	Cohen's d [†]
1st person singular (I)	t-test	10.863	1.544	9.730	2.301	1.910	0.0630	0.4550	-0.578
Personal pronouns	t-test	14.113	1.609	13.068	2.566	1.619	0.1130	0.4550	-0.488
2nd person (you)	Mann-Whitney U	1.016	0.639	1.408	0.998	157.000	0.1370	0.4550	0.468
Absolutist words	t-test	0.989	0.429	0.792	0.437	1.457	0.1530	0.4550	-0.454
Pronouns (total)	t-test	20.503	1.681	19.448	3.485	1.298	0.2020	0.4550	-0.385
Positive emotion	Mann-Whitney U	2.022	0.729	1.719	0.964	270.000	0.1740	0.4550	-0.355
Positive tone	Mann-Whitney U	4.360	1.030	3.945	1.317	264.000	0.2270	0.4550	-0.351
3rd person singular (she/he)	Mann-Whitney U	0.829	0.622	0.656	0.494	250.000	0.3940	0.6250	-0.309
Sadness	t-test	0.309	0.213	0.251	0.231	0.837	0.4080	0.6250	-0.259
1st person plural (we)	Mann-Whitney U	0.499	0.419	0.393	0.424	275.500	0.1330	0.4550	-0.250
Negative emotion	Mann-Whitney U	1.462	0.763	1.603	0.880	201.500	0.7220	0.8190	0.171
Anxiety	Mann-Whitney U	0.252	0.192	0.219	0.213	247.000	0.4380	0.6250	-0.161
Negative tone	Mann-Whitney U	2.200	0.968	2.326	0.766	187.000	0.4690	0.6250	0.144
Anger	Mann-Whitney U	0.429	0.450	0.384	0.378	232.500	0.6840	0.8190	-0.108
3rd person plural (they)	t-test	0.767	0.495	0.721	0.513	0.297	0.7680	0.8190	-0.092
Impersonal pronouns	t-test	6.389	0.735	6.379	1.801	0.023	0.9820	0.9820	-0.007

[†] Positive Cohen's d indicates higher scores in depressed men; negative values indicate higher scores in depressed women.

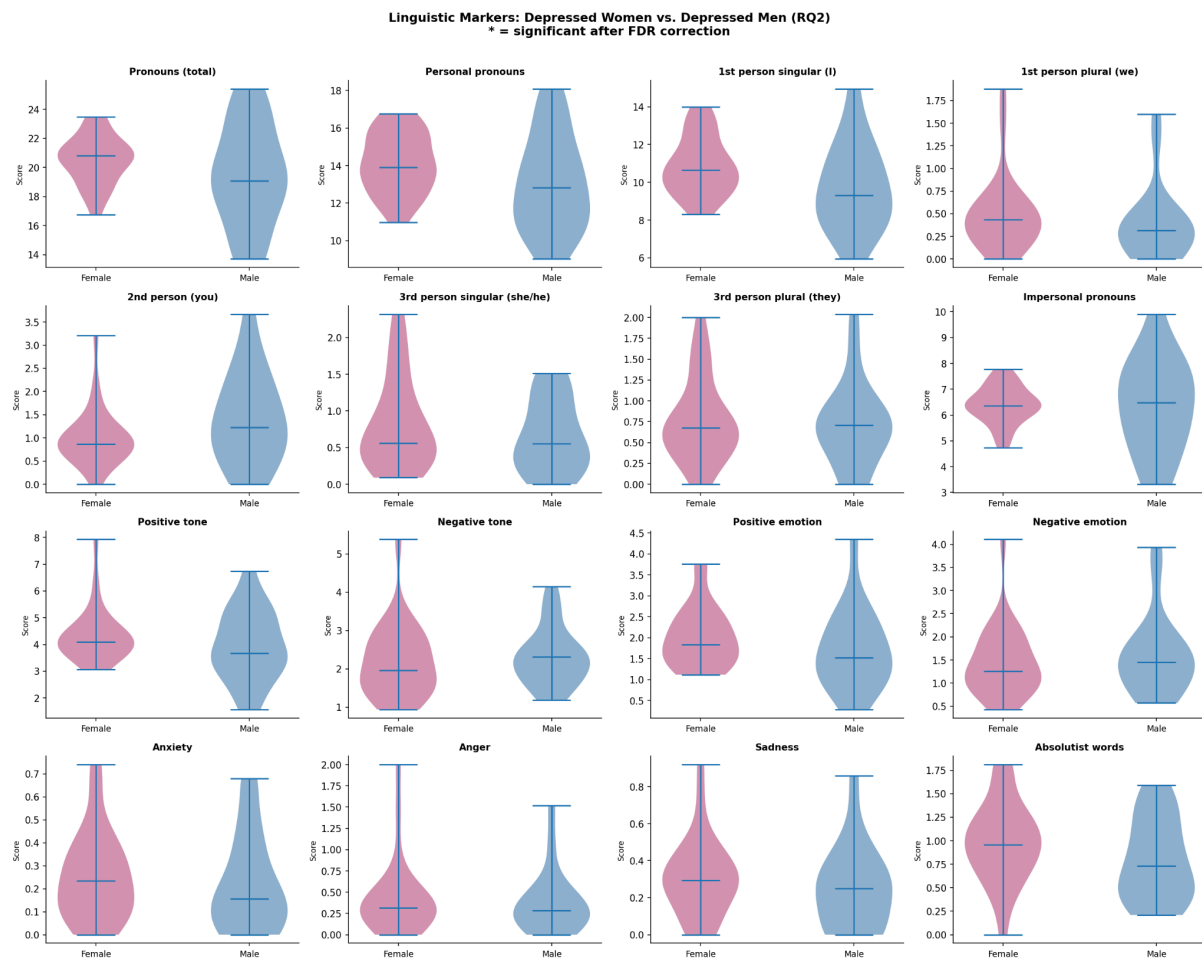


Figure 3. Violin plots presenting the distribution of differences between depressed women (pink) and depressed men (blue) per linguistic marker. No markers reached statistical significance.

To approach Hypothesis 2 from a different perspective, two logistic regression models were run on the data. However, as independent samples analyses showed lack of significance between male and female depressed samples, the two groups were analysed separately. Therefore, the models were not directly testing how depressed men differ from depressed women in terms of linguistic markers. Instead, they were testing which linguistic markers are significant cues of depression in women, and which are significant cues in men. This methodology allows for a more descriptive analysis of depression profiles between male and female participants. The logistic regression for the female sample demonstrated modest discriminative ability, with a cross-validated AUC value of 0.681 (SD = ± 0.126), a macro-averaged F1 score of 0.77, and a weighted average F1 score of 0.79. Similarly to the full sample model for Hypothesis 1 (Figure 2.), the model was much better at predicting non-depressed women (recall = 0.90) than depressed women (recall = 0.62). While some predictors reached statistical significance, due to the model's modest predictive power, its

classificatory ability may be limited. The performance of the male-only logistic regression model was considerably weaker. The model's cross-validated AUC was equal to 0.526 (SD = ± 0.228), making its predictive ability no better than chance level, which likely reflects the small sample size of depressed men (N = 18). The results will be presented, yet should only be interpreted as possible trends, and not statistically significant findings.

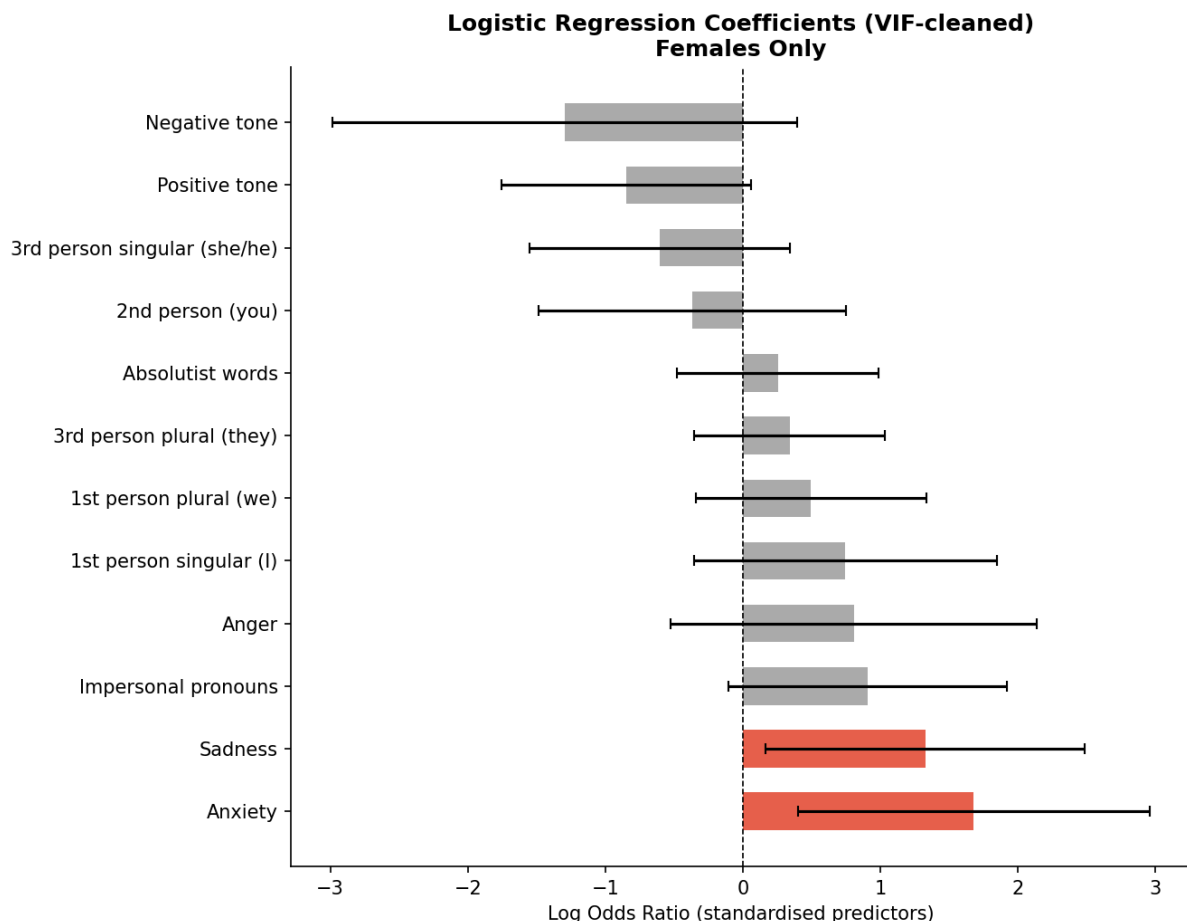


Figure 4. Logistic regression coefficients for linguistic markers (n = 12) when predicting from the female sample (n = 63). Log odds ratios are presented with 95% confidence intervals. Positive values indicate a higher risk of depression, while negative values indicate lower risk. Statistically significant markers ($p < 0.05$) are highlighted in red. Predictors are standardised.

According to the model, two linguistic markers were statistically significant predictors of depression in the female sample: Anxiety (OR = 5.36, 95% CI [1.49, 19.29], $p = 0.01$) and Sadness (OR = 3.77, 95% CI [1.18, 12.04], $p = 0.025$). While both predictors reached significance, the wide confidence intervals reflect the limited sample size, and suggest that interpretations based on the results should be treated with caution. The odds ratio of the Anxiety marker in depressed women (OR = 5.36) is particularly elevated in comparison to the

odds ratio of the full sample ($OR = 1.98$), which suggests that anxious language is a particularly salient linguistic marker of depression for women specifically. The second significant marker was sad emotive language, which depressed women exhibited approximately five times higher odds of using than non-depressed women. This is notable as sadness did not reach significance in the full sample model, suggesting it may be a gender-specific marker of depression. While not significant statistically, the Negative Emotion marker was a negative predictor of depression, similarly to the full sample; this is consistent with the previous observation that when all markers are considered simultaneously, negative emotion words are not reliable predictors of depressive symptoms.

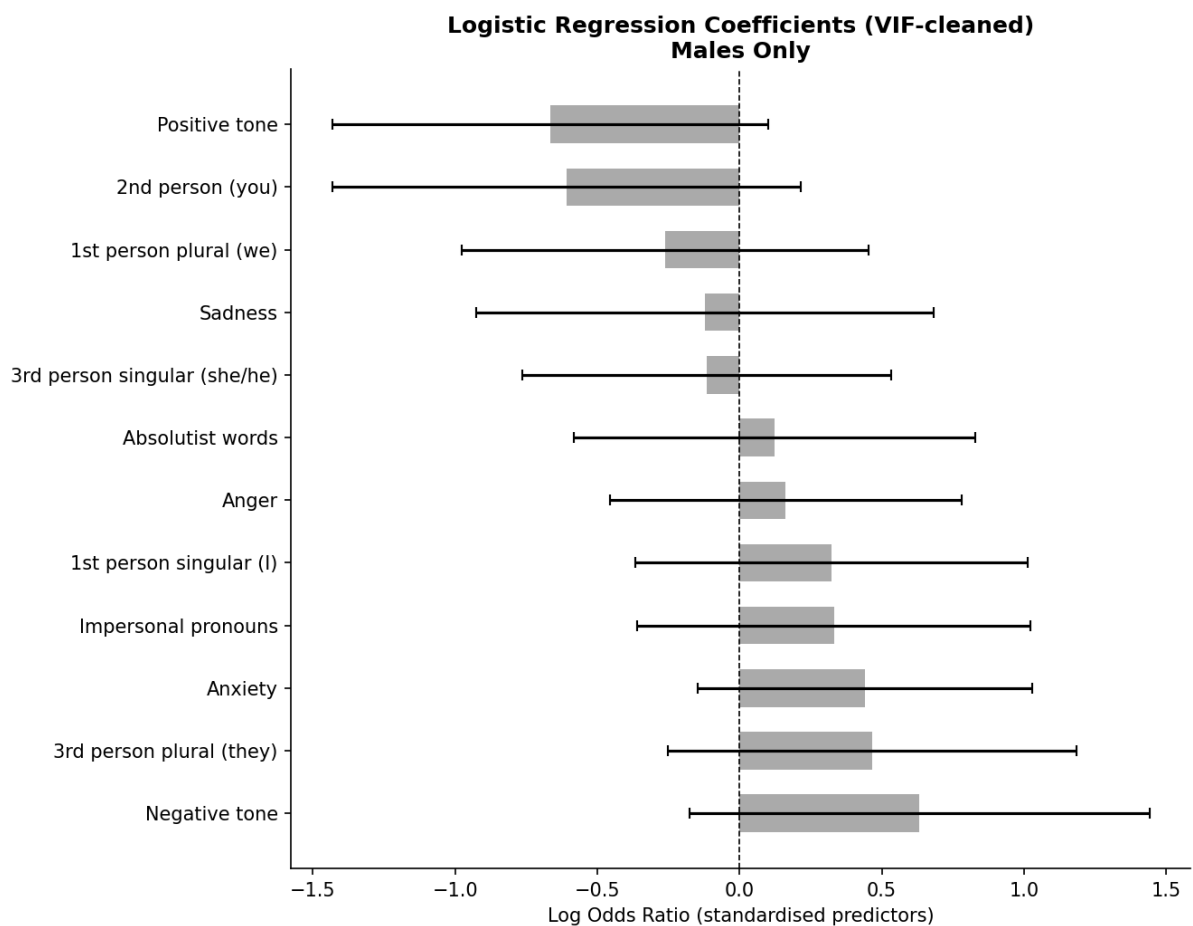


Figure 5. Logistic regression coefficients for linguistic markers ($n = 12$) when predicting from the male sample ($n = 79$). Log odds ratios are presented with 95% confidence intervals. Positive values indicate a higher risk of depression, while negative values indicate lower risk. Statistically significant markers ($p < 0.05$) are highlighted in red. Predictors are standardised.

As reported above, the male-only sample model did not find any statistically significant predictors of depression in linguistic markers, which is consistent with its weak performance in the classification task; this is most likely caused by the small sample size. For these reasons, all inferences based on this model's predictions are only possible trends within the data, and not statistically significant findings. The trends within the data align with findings from previous literature; strong positive markers of depression were Negative Tone (OR = 1.89, $p = 0.13$), Anxiety (OR = 1.55, $p = 0.141$), and FPSP (OR = 1.39, $p = 0.35$). Sadness words, which were strong positive predictors in the female sample, were negatively directed in the male sample. No other markers showed notable directional differences between the two models. For more robust results, the analysis would require repeating on a larger sample.

Taking both independent samples and logistic regression analyses into consideration, Hypothesis 2 cannot be confirmed; within the analysed dataset, depressed women and depressed men do not differ significantly in terms of language use. The independent samples analysis reported no significant differences between the two samples per linguistic marker. Complementary logistic regression models were conducted on each sample; the female sample exhibited a distinct linguistic profile, with Sadness and Anxiety being significant positive markers of depression, while the male sample exhibited no statistically significant markers. These findings may suggest that the linguistic expression of depression may differ between men and women, however, it is important to note that the models exhibited weak-to-modest predictive power. Further analyses on larger data samples would need to be conducted to re-test Hypothesis 2.

2. Discussion

Upon analysis, Hypothesis 1 was partially confirmed. At the level of the sub-hypotheses, Hypothesis 1a received partial support through the significance of Anxiety, Negative Emotion, and Negative Tone markers. Hypotheses 1b and 1c were not confirmed. Independent tests revealed that Anxiety, Negative Emotion and Negative Tone were statistically significant cues, being more prevalent in the depressed sample. A logistic regression model confirmed the Anxiety marker finding, however did not find the Negative Tone marker to be significant; in turn, the logistic regression showed that the Second Person "You" Pronoun and Positive Tone markers were negatively correlated with depression. Hypothesis 2 was not confirmed; independent samples analyses did not find any significant linguistic marker differences between male and female depressed participants. Separate logistic regression models were run on the male-only sample and the female-only sample, to

construct linguistic profiles of depressed men and depressed women in comparison to their healthy counterparts. In women, Anxiety and Sadness markers are significant cues of depression. In men, no markers were found to be significant. While not all markers were found to be significant across the two research questions, each of four groups (depressed women, non-depressed women, depressed men, and non-depressed men) exhibited a characteristic linguistic profile (see Figure 6.).

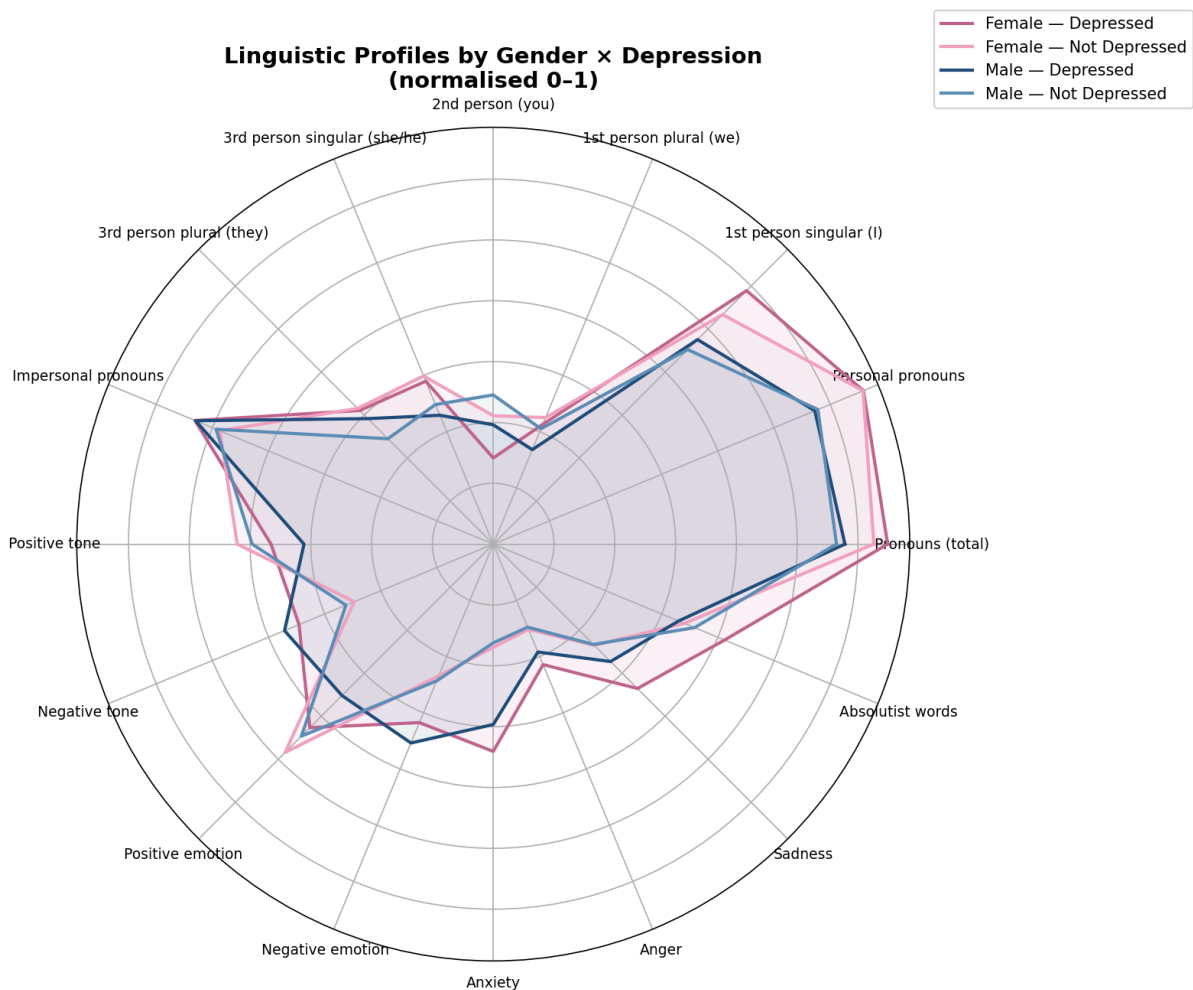


Figure 6. Radar chart depicting the normalised mean linguistic marker profiles of depressed and non-depressed men and women across 16 LIWC-derived markers (min-max normalised, 0–1). The overall similarity of profiles across groups reflects the modest group differences observed in the statistical analyses.

The three markers found to be significant for RQ1—Anxiety, Negative Tone, and Negative Emotion—are in line with previous literature on linguistic markers of depression. As concluded by Savekar et al. (2023) in their systematic literature review, depressed mood becomes apparent in the form of negatively-charged language use; speaking with a negative tone can also be linked to what the authors call “emotional dysregulation and frequent

mood-switching” (2023, pp. 213). This is consistent with the findings of the present study. Moreover, the findings regarding Negative Tone ($d = 0.602$) strongly align with the ideas posed within Beck’s (1967) Cognitive Model of Depression, in which the author states that negative affect is a strong indicator of a depressed mental state; as stated by Tausczik and Pennebaker (2009), and further explicated in this study, this negative cognitive state manifests itself in a patient’s linguistic behaviour.

The Anxiety marker is of particular significance, as it was robust across both t-tests and logistic regression models. Crucially, it was the sole marker to do so; this convergence across methodologically distinct approaches constitutes the most robust evidence produced by the present study, as it demonstrates that the association between anxious language and depression is not an artefact of any single analytical method. It is possible that anxiety is a further form of emotional dysregulation (Savekar et al., 2022), which is very often present in depressed patients; for instance, the ICD-11 specifications (see Chapter One, Section 4.2.2) distinctly present the depression diagnosis modifier “with prominent anxiety symptoms (6A80.0)” (icd.who.int/browse/2025-01/mms/en#1194756772; DOA 06.02.2026). However, previous studies do not detail anxious language as a significant aspect of depressed language, instead focusing on negative affect and pronoun use (for an exception, see: Stade et al., 2023); Stade et al. (2023) specifically found that depression and anxiety share overlapping linguistic patterns, with anxious language present in both conditions. This convergence may explain why anxious language is particularly salient in depressed individuals; if depression and anxiety are clinically comorbid conditions that share linguistic expression, as the ICD-11 modifier system acknowledges, then anxious language may be among the most reliable detectable features of depression in spoken data. Thus, the findings of the present study may imply that more attention must be placed on the Anxiety marker in future research.

Within the negative emotion category, the Sadness marker deserves consideration; although it did not survive the later FDR correction, Sadness was a significant marker upon initial testing ($p_{\text{raw}} = 0.049$). This finding partially aligns with previous research on linguistic markers across mental disorders, in which studies indicate that Sadness is most relevant in MDD, when compared to schizophrenia, borderline personality disorder, generalised anxiety disorder, and others (Lyons et al., 2018). A further noteworthy finding from the logistic regression was the negative association between Positive Tone and depression ($OR = 0.53$, $p = 0.02$), indicating that higher use of positively valenced language was associated with lower odds of depression. While Positive Tone did not reach significance in the initial t-tests, this result is theoretically consistent with Beck’s (1967) cognitive model,

which predicts not only elevated negative affect but also the suppression of positive affect as a feature of depressive cognition. The finding suggests that depressed language is characterised not only by what is present, i.e. negative and anxious language, but also by what is absent: positively toned expression. Taken together, the findings suggest that a depressed individual's cognitive state is characterised by increased negative emotions, which later manifest themselves in negative, specifically anxious, language, and a decrease in positive affect.. On the whole, these markers can be considered strong indicators of depressed mood, serving as strong candidates for linguistically-based diagnostic tools.

Other markers, however, were not replicated; this applies to the markers targeted by Hypotheses 1b (First Person Singular Pronoun (FPSP)) and 1c (Absolutist Words), which were strongly grounded in literature. In terms of FPSP, the failure to replicate previous findings may be attributed to many factors. Tackman et al. (2019) reported elevated FPSP use in depressed individuals, even when controlling for gender; however, this study was conducted on written discourse. The DAIC-WOZ is a corpus of spoken interviews, which may suggest a modality effect, meaning that FPSP is more consistently elevated in written text, and not in spoken language. This may be due to the nature of spoken dialogue; speech has a tendency to be fragmented, which was also the case in the current corpus, with participants often replying in single, highly pragmatic utterances instead of full sentences (e.g. “Uh, my father”, see Chapter Two, Section 1.2). In such a scenario, FPSP may innately be less present, reducing its detectability as a linguistic marker of depression. Similarly, Trifu et al. (2017) reported the FPSP effect when analysing a narrative elicitation task, which is also qualitatively different from the spoken process of asking and answering questions. Another explanation for the lack of FPSP significance in the present study may be rooted in the methodological tool. While LIWC is a well-established tool for quantitative linguistic analyses, it may not be suited for noticing contextual cues (Savekar et al., 2022). Moreover, the LIWC user manual highlights that analysis quality increases proportionally to the text word-count; the short and fragmented answers provided by participants within the DAIC-WOZ may not provide sufficient linguistic material for catching the FPSP marker effect during analysis. Other explanations may be attributed to the structure of LIWC dictionaries; Zimmerman et al. (2017) found that objective and possessive FPSP use (“me”, “my”) was more significant statistically among depressed participants than the subjective “I” FPSP. The LIWC dictionary specifically targets the “I” pronoun, meaning that certain FPSP effects were overlooked in the analysis.

A similar non-replication pattern was observed in terms of absolutist words, which are a linguistic marker of depression identified by Al-Mosaiwi and Johnstone (2018). Possible explanations are similar to those posed for FPSP use; previous research was conducted on online forums, which are distinctive linguistically, due to their written, asynchronous and anonymous nature. It may be that absolutist words are common in the written discourse of depressed individuals when they are writing in safe settings, protected by anonymity, but less salient in more stressful, impromptu interview settings. An alternative explanation is linked to the nature of the PHQ-8 psychological test; the original study found that absolutist words were most prominent in suicidal ideation forums. The PHQ-8 approach adopted by the DAIC-WOZ provides a binary cut-off point, where ≥ 10 points equals a positive depression diagnosis. Without controlling for depression severity, certain markers, such as absolutist words, may be less justifiable as diagnostic cues.

The DAIC-WOZ corpus introduces a further set of methodological considerations that may have shaped the present results. The interviews were conducted by the Ellie agent, a virtual computer avatar, meaning that it is not guaranteed that participants behaved as they would in a naturalistic conversational context. It is possible that certain significant markers, such as Anxiety, were elevated due to the unnatural interview setting, while others previously well-documented in human-human communication, such as FPSP, may have been suppressed or over-compensated for. Additionally, halfway through the data collection, two framing conditions were introduced: one in which participants knew that a human was controlling the agent, and one in which they were uninformed. As the corpus does not record which condition each participant was assigned to, an uncontrolled framing effect cannot be ruled out. This is of particular relevance, as Gratch et al. (2014, pp. 3127) report that “participants also displayed sad emotional expressions more intensely when they believed they were interacting with a computer compared to a human”, and that “participants felt lower fear of negative evaluation and engaged in less impression management when the character was framed as autonomous than when it was framed as human-controlled”. Together, these corpus-specific factors may have systematically altered participants’ linguistic behaviour in ways that are difficult to disentangle from the genuine effects of depression on language.

The significant Second Person “You” Pronoun (SPP) logistic regression result reported in this study (OR = 0.54, 95% CI [0.29, 0.98], $p = .042$) is a relatively unexplored finding in research on the linguistic markers of depression. The marker also came close to significance in the initial t-testing ($p_{\text{raw}} = 0.072$), but later failed to survive FDR correction ($p_{\text{FDR}} = 0.165$). Very few studies exist which have previously found such trends in the

language of depressed people; Leis et al. (2019) reported that, when looking specifically at pronoun use in depressed language, “the first- and the second-person plural pronouns were the least frequent (0.4% in both cases), this distribution being different from that of the control dataset ($P < .001$)” (pp. 1). However, the study was conducted on Spanish tweets, which differ from the DAIC-WOZ in terms of modality and language; these aspects may be significant, as researchers warn that markers of depression may not be generalisable across cultures and languages (Savekar et al., 2022). Despite limited empirical precedent, this finding is backed by theoretical explanations of depressive behaviours. Jarrold et al. (2011) proposed the “self-focused theory of language in depression”, which suggests that depressed individuals withdraw inward and possess a tendency to self-ruminate. If taken to be true, focusing on the self may cause depressed individuals to focus less on the “other”, in turn resulting in lowered SPP use. The finding of the present study may suggest that the inverse relationship between the SPP pronoun and depressed language requires equal attention to the FPSP marker.

In terms of RQ2, the independent samples analyses did not reveal any significant difference between linguistic marker use in depressed men and depressed women. This null result should, however, be treated with caution; due to the size of the available DAIC-WOZ dataset, upon removing non-depressed participants, the remaining depressed subsample was considerably small ($N = 42$). The analysis was certainly underpowered, considering that the two small groups of depressed women ($N = 24$) and depressed men ($N = 18$) were compared across 16 variables. Moreover, Parker and Brotchie (2010) noted that gender differences are only predominant during the first episode of depression; if certain participants in the database suffered from chronic or recurrent depression, the relevance of gender may simply be concealed among the results. While it is possible that this study’s results suggest an absence of differences between language use in depressed men and women, it may also be a case of absence of evidence. Further research, conducted on larger and more controlled samples, is required.

Despite the lack of statistical significance when comparing depressed men and depressed women, separate logistic regression models were applied to subsamples of all men and all women, predicting depression status within each group independently. The results should be interpreted as loose, exploratory trends, due to the models’ modest discriminative abilities, however, they can still show insights into the gender-specific linguistic differences. In the female subsample, two markers were significantly elevated: Anxiety ($OR = 5.36$, 95%

CI [1.49, 19.29], $p = 0.01$) and Sadness (OR = 3.77, 95% CI [1.18, 12.04], $p = 0.025$). The Anxiety finding corresponds to previous medical literature, which noted increased predisposition to anxiety in women rather than men (Marcus et al., 2005); official manuals, such as the ICD-11 guidelines, state that “women with a diagnosis of a Depressive Disorder are more likely to experience co-occurring Anxiety or Fear-Related Disorders”. A developmental explanation for such behaviour is girls’ tendency to develop anxious coping styles in early-to-mid childhood (Carter, 2011); such behaviour may later be extrapolated to language. This finding suggests that Anxiety may be a more effective linguistic marker of depression within female samples, rather than within male samples. Furthermore, the substantially elevated odds ratio for anxiety in the female model (OR = 5.36) compared to the full sample model (OR = 2.03) suggests that the full sample effect may be largely driven by the female subsample. This raises the possibility that anxiety language is primarily a marker of depression in women specifically, and that its appearance as significant in the full sample reflects women’s disproportionate contribution to the signal rather than a universal pattern across both genders. This interpretation is consistent with the ICD-11’s recognition of anxiety as a gendered comorbidity of depression, and with developmental accounts of anxious coping in women (Carter, 2011 in Altemus et al., 2014).

Sadness as a gender-specific marker can also be placed within previous literature; for instance, Altemus et al. (2014) stated that women are more likely to experience interpersonal sensitivity, which can result in increased feelings of sadness and exclusion. Parker and Brotchie (2010) also highlighted that women are more likely to internalize, leaning towards behaviours such as crying or self-isolating. In light of these observations, Sadness may be a more reliable indicator of depression in women, than in men. The implications of these two findings for the Silencing the Self theory (see Chapter One, Section 6.2) are partially contradictory. On the one hand, Jack (1991) suggested that women are more likely to suppress negative thoughts and emotions; in line with this observation, it would be expected that Sadness and Anxiety markers would not be elevated among women with depression, but instead either neutral, or even reduced. As the Sadness marker was elevated, it may be possible that the negative cognitive effects of depression override the significance of self-silencing, presenting themselves in linguistic contexts during symptom-specific patient interviews. On the other hand, Jack (2010) also argued for the existence of “the divided self” in women, a phenomenon characterised by “dissonance between inward experience and outside presence” (pp. 6). Sadness-related language may be a manifestation of the inward experience “breaking out” into outward expression, in line with the STS theory assumptions.

Importantly, Jack (1991) underscored the particular significance of women's anger suppression; the Anger marker result in the present study ($OR = 2.25, p = 0.233$) is broadly consistent with these observations, as depressed women did not exhibit elevated anger-related language use. Taken together, the female-specific markers of depression portray the complicated nature of embedding empirical findings within existing theoretical models; in the diagnostic context, there may be reason to believe that social aspects, such as gender, impact linguistic profile (Figure 6).

As with the female model, the results and interpretations of the male linguistic profiling should be treated as exploratory analyses. In contrast to the female sample, no linguistic markers of depression reached statistical significance in the logistic regression model run on male participants. As explained with regard to the independent samples test result, this may be the consequence of small depressed samples ($N_{dep.} = 18, N_{non-dep.} = 61$). Nonetheless, the strongest positive male trend was Negative Tone ($OR = 1.88$), which is consistent with Beck's (1967) Cognitive Model of Depression. While little literature exists on male-specific experiences of depression, some sources note that men exhibit a higher tendency to externalize and show anger (Parker and Brotchie, 2010), and are more prone to substance abuse (Marcus et al., 2005). However, these observations are not reflected in the present study's results, as depressed men did not exhibit significantly higher anger-related language use than non-depressed men, or depressed women. It is possible that such externalising behaviours are not expressed linguistically. Alternatively, the STS theory (Jack, 1991) is specific to women, and few studies have analysed depressed men from the perspective of linguistics. Levant (2025) proposed the Normative Male Alexithymia hypothesis, which claims that men "are taught by family, peers, and society to suppress vulnerable emotions for fear of being perceived as unmasculine" (pp. 3). In light of these findings, it is possible that male participants also suppressed emotional behaviours linguistically. As broader implications, such claims suggest that the phenomenon of self-silencing is experienced by both men and women, and deserves further consideration in future research on not only depression and language, but gender and emotion as a whole.

When analysed holistically, the results of the present study carry implications for the tasks of automating diagnostic tools, linguistic marker analyses, and gender-sensitive depression detection. Regarding automated diagnostic tools, the study was framed from the perspective of Affective Computing, an interdisciplinary field tasked with programming computers to understand and to exhibit emotion and emotional intelligence (Picard, 1997). The field is directly linked to the topic of depression and mental health, as presently,

researchers are exploring possibilities of automating depression detection with the help of automated computer agents or AI chatbots (Schlicher et al., 2025; Ren et al., 2025); a prime example of this is the DAIC-WOZ corpus used within the present study, which was constructed with the hope of aiding automatic symptom detection, and employed the “Ellie” agent as a tool for conducting interviews (Gratch et al., 2014; DeVault et al., 2014). The results of the present study can be interpreted as promising with regard to the current state of AffC in mental health research. The study yielded some statistically significant results, suggesting that quantitative analyses of language can be used as an aid during the diagnostic process of depression. Furthermore, the robustness of the Anxiety marker across analytical approaches and its particular salience in the female subsample suggest it should be prioritised as a candidate feature in linguistically-based depression screening tools. Future tools designed with gender-sensitivity in mind could weight anxiety language more heavily when screening women, while investigating whether alternative markers, such as negative tone, may be more informative for men. However, for more reliable and valid integration of automatic AffC tools in the medical field, considerable comparative research must be done, paying attention to possible linguistic and behavioural differences between human-human and human-computer interactions.

The study also raises questions about the sufficiency of linguistic markers for depression detection. While the results of this study indicate that significant traces of depressed cognitive state appear in language, the markers may not be robust enough to serve as stand-alone diagnostic tools, especially without further qualitative analysis which could infer information from context. Although the study revealed significant linguistic markers of depression, such as Anxiety or Negative Emotion, other markers which were well-documented in prior literature were not replicated; this suggests that quantitative linguistic analyses should be integrated into the diagnostic process as part of a multimodal framework, and not as decisive, freestanding indicators of depression. Park and Park (2018) noted that non-verbal aspects make up approximately 90% of successful communication (see Chapter One, Section 5.1), while Hannigan et al. (2020) comment that distressed individuals, such as people dealing with depression, often communicate even more insightful information in a paraverbal manner. With this in mind, text-only analysis possibly catches the remaining 10% of signals, missing cues which may have been more prominent in facial expressions (Girard et al., 2013), body movements (Fiquer et al., 2018), gestures, or vocal quality (Seo et al., 2024). The use of linguistic markers of depression should, therefore, be used alongside other analyses whenever possible, in order to create holistic and effective diagnostic tools;

LIWC-based linguistic results could be accompanied by the analysis of paralinguistic features like speech rate, as well as insights from facial and bodily recognition software, to provide for more reliable diagnoses. As the intertwined nature of language, cognition, and mental health state is complex and multifaceted, future computational tools must be programmed with this in mind.

In terms of the gendered linguistic aspects of depression, the independent samples tests reported no differences between depressed men and women, yet logistic regression did find the two groups to exhibit diverse linguistic profiles. Differences between male and female experiences of depression have been reported in many studies (see Chapter One, Section 6.1), and are explicitly listed in the ICD-11 manual (icd.who.int/browse/2025-01/mms/en#1563440232, DOA 04.02.2026). Therefore, it is possible that although these symptomatic differences exist, they do not manifest themselves linguistically to the extent of being distinguishable between genders. However, in light of the relatively small data sample in the present study, it is important to consider the existence of linguistic differences in future research; when summarising the results of their text-based depression analysis study, Havigerova et al. (2019) concluded that “it provides clear that models predicting the depression differ depending on gender, and that in the future it is necessary to take into account the moderative/mediatory influence of the writer/speaker’s gender in modeling relationships between depression (or other personality characteristics) and text” (pp. 8). As literature on the topic is sparse, it is clear that more research must be done in order to replicate results such as Havigerova and colleagues’, or to reassess the question posed within the present study. The development of automated tools which do not account for potential gender differences, creates potential risk for groups being mis- or under-represented, deepening social inequalities: a concern which resonates with the neurofeminist critical approach towards gendered biases in clinical research (see Chapter One, Section 6).

3. Limitations

The study was conducted so as to limit any possible confounding variables, so as to provide reliable findings; however, not all factors could be controlled for, resulting in possible limitations. The primary limitations may be attributed to the chosen source corpus. The Distress Analysis Interview Corpus (WOZ version) is one of the leading corpora for analysis of depression and PTSD (Gratch et al., 2014; DeVault et al., 2014). It has been employed in analyses of linguistic markers, such as in the present study, as well as in studies of vocal (Huang et al., 2024; Pratiwi and Hayati, 2025) and facial (Tiwary et al., 2023; Aghaei and

Khodaei, 2023) cues. Despite its widespread use, certain characteristics of the corpus posed potential issues. The sample size of the corpus may have influenced the results in two ways; due to the relatively small number of depressed participants, certain markers may have been overlooked, and due to the proportional imbalance between depressed and non-depressed individuals, any statistical models may have been skewed in favour of the majority group. Apart from the sample size, certain procedural decisions may have also influenced the data: for instance, partway through data collection, overt and covert conditions were introduced. Specifically, certain participants were made aware of the fact that a human was behind the computer agent, while others were made to believe that the agent was fully automated. The data regarding the participants' distribution for this condition was not available within the corpus, meaning that this condition could not be controlled for. The implications of this experimental framing may have relevance for the applications of affective computing in clinical contexts (cf. Gratch et al., 2014), however, cannot be explicitly discussed in light of the present study. Moreover, the corpus treated gender and depression status as categorical binaries. Such a reductionalist approach ignores the complex and intersectional aspects of not only gender (Duchesne, 2025), but also the spectrum of depression. In terms of the latter, the PHQ-8 psychological test which was used within the corpus was constructed according to DSM-IV guidelines, meaning that it may not be up to present DSM-5 diagnostic standards. The test also flattens the depression continuum presented within the DSM-5 manual by presenting participants with scores ≥ 10 as depressed, potentially grouping together cases of mild and severe depression. Such simplifications of complex matters may have concealed the linguistic signs of depression.

Apart from corpus-related restraints, certain limitations linked to data generalisability exist. The linguistic analyses were conducted on transcripts of spoken interviews; a systematic literature review by Savekar et al. (2022) presents notable fluctuation in linguistic markers of depression between modalities, and Park and Park (2018) conclude that any linguistic modality is insufficient, if paralinguistic and non-verbal aspects are not taken into consideration. The analysis conducted within the present study should be treated as an insight into a fraction of a complex scientific issue, that being the detection of depression. Importantly, the analysis was conducted on data provided by English-speaking American citizens; Savekar et al. (2022) underscore the fact that linguistic markers may differ between cultures and languages, while Summerfield (2008) highlights the Western origin of the DSM-5 and ICD-11 diagnostic criteria. It is possible that different linguistic markers of depression would be identified in corpora containing data from non-WEIRD societies.

4. Future Directions

As Affective Computing is gaining momentum and branching into clinical scientific studies, further research on the interplay of depression and language, as well as gender, is necessary. Future studies would benefit from analysing larger, gender-balanced samples, so as to provide more robust results, ensuring adequate statistical power. Datasets could employ wider sets of depression questionnaires, and control for depression episode occurrence, as Parker and Brotchie (2010) highlight that gender differences are the most prominent in terms of first onset episode. Cross-linguistic replication would allow for better modeling of depression worldwide, avoiding Western biases, both in terms of language use and depression experience. In the future, linguistic analyses should be integrated within larger multimodal frameworks, including paralinguistic and non-verbal features, so as to provide the most reliable solutions for automated depression detection software. Finally, more research is required on the effects of gender on the use of linguistic markers of depression; as Schlicher et al. (2025) highlight that AffC is moving towards creating models which cater to personal needs, the important role of gendered experiences should be explicitly incorporated into model design. Any clinical tool should aim for inclusivity and lack of bias, so as to avoid the exclusion of underrepresented minorities, such as women, non-Western populations, and gender-diverse individuals.

Conclusion

The aim of this MA thesis was to verify whether depressed and non-depressed individuals use language differently by comparing a set of empirically grounded linguistic markers (Negative Valence, First Person Singular Pronoun, Absolutist Words), and to determine whether these linguistic differences are independent of an individual's gender/sex. In terms of the former, I hypothesised that depressed individuals would use more negatively valenced language, first person singular pronouns, and absolutist language when compared to non-depressed controls. For the latter, I hypothesised that these linguistic cues of depression would not be equally prevalent in men and women. To test these hypotheses, a corpus-based Linguistic Inquiry and Word Count (LIWC) analysis of spoken interview transcripts from the Distress Analysis Interview Corpus – Wizard of Oz (DAIC-WOZ) corpus was conducted. My results were partially statistically significant.

In Chapter One I provided the theoretical background of this interdisciplinary study. The chapter describes the fields of Natural Language Processing, Affective Computing, and Mental Health research. It continues with an in-depth presentation of depression, and identifies links between gender/sex, mental health, and language, specifically targeting the Silencing the Self theory. The chapter concludes with an analysis of the presented issues through the lens of neurofeminist research.

In Chapter Two I presented the methodology behind the study. The chapter describes the DAIC-WOZ corpus and details the posed research questions, alongside the related hypotheses. It also explains the functioning of the LIWC software, provides the theoretical and empirical background of the selected linguistic markers of depression, and describes the process of constructing the final database.

In Chapter Three I presented and discussed the results of the conducted analyses. With regard to the first group of hypotheses, the results indicate that depressed and non-depressed people differ in terms of linguistic marker use. The markers found to be of relevance were Anxiety, Negative Tone, and Negative Emotion. Surprisingly, markers which have previously been identified in prior literature, the First Person Singular Pronoun and Absolutist Words, were not replicated within the present study. These results suggest that linguistic markers of depression may be dependent on many additional factors, such as text modality (spoken vs. written) or text source (narrative elicitation tasks, interviews, forum posts). While linguistic markers may serve as useful cues in the task of depression detection, they would benefit from being integrated into wider frameworks, which additionally consider facial expression, body language, and paralinguistic features.

The second hypothesis was not confirmed. Inferential group comparison analyses of the linguistic markers used by male and female depressed participants did not present significant differences. Further purely exploratory logistic regression analyses were conducted to present the specific linguistic profiles of depressed men and depressed women, when compared to their healthy counterparts. Notably, depressed women exhibited high Anxiety and Sadness marker scores, while the language of depressed men did not differ significantly from that of non-depressed men. These exploratory results suggest that subtle gender differences may emerge, and that the effects of gender/sex require further study in clinical Affective Computing research concerned with depression detection.

The study was primarily subject to limitations linked to relatively small and unbalanced sample sizes, as well as to generalisability issues caused by the specificity of the sample (i.e. WEIRD population, spoken interviews, human-computer interaction). For these reasons, I believe that the study would benefit from being conducted once again. If given the opportunity, I would repeat the study on a larger sample balanced between both depressed and non-depressed participants, as well as men and women. I would consider using continuous operationalisations of depression, and non-binary operationalisations of gender/sex, allowing for more complex analyses. However, as of now, the results of this MA thesis suggest that: depressed and non-depressed participants differ in terms of specific linguistic marker use; no significant gender effects occur in terms of these linguistic markers of depression. While relevant gender differences have been identified in theoretical and clinical research, they may not manifest linguistically, unless proven otherwise in future research.

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Streszczenie

Niniejsza praca magisterska bada związki między depresją, językiem i płcią, koncentrując się na pytaniu, czy osoby z depresją i bez depresji różnią się pod względem użycia określonych markerów językowych, oraz czy ewentualne różnice są zależne od płci badanych. Analizy przeprowadzono przy użyciu narzędzia LIWC na transkryptach wywiadów z korpusu DAIC-WOZ, obejmującego nagrania półustrukturyzowanych rozmów prowadzonych przez wirtualnego agenta konwersacyjnego.

Podstawę teoretyczną pracy stanowią zagadnienia z zakresu przetwarzania języka naturalnego, informatyki afektywnej oraz psychiatrii klinicznej, ze szczególnym uwzględnieniem różnic między płciami w przebiegu i objawach depresji. Kluczową rolę odgrywa teoria *Silencing the Self*, opisująca mechanizmy tłumienia emocji u kobiet, oraz perspektywa neurofeministyczna, kwestionująca uproszczone podejścia do kategorii płci w badaniach naukowych.

Wyniki analiz są częściowo zgodne z postawionymi hipotezami. W odniesieniu do pierwszego pytania badawczego wykazano, że język osób z depresją charakteryzuje się wyższym poziomem Lęku (*ang.* Anxiety), Negatywnego Tonu oraz Negatywnych Emocji. Nie potwierdzono natomiast obecności markerów dobrze udokumentowanych w dotychczasowej literaturze, tzn. zaimków pierwszej osoby liczby pojedynczej oraz słów absolutystycznych, co może wynikać z odmiennej modalności analizowanego materiału w porównaniu z wcześniejszymi badaniami opartymi głównie na tekstach pisanych. Druga hipoteza nie została potwierdzona; analiza nie wykazała istotnych różnic między językiem kobiet z depresją a mężczyzn z depresją. Eksploracyjne analizy regresji logistycznej ujawniły jednak, że u kobiet z depresją istotnymi predyktorami były markery Lęku i Smutku, podczas gdy w grupie mężczyzn żaden marker nie osiągnął istotności statystycznej.

Wyniki sugerują, że językowe markery depresji stanowią obiecujące narzędzie wspomagające diagnozę, lecz wymagają integracji z szerszymi ramami multimodalnymi oraz weryfikacji na większych i bardziej zróżnicowanych próbach. Kwestia różnic płciowych w językowym wyrażaniu depresji pozostaje otwarta i wymaga dalszych badań.